



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

Operating Systems

Lecture 3

Threads and processes: a quick, programmer's viewpoint

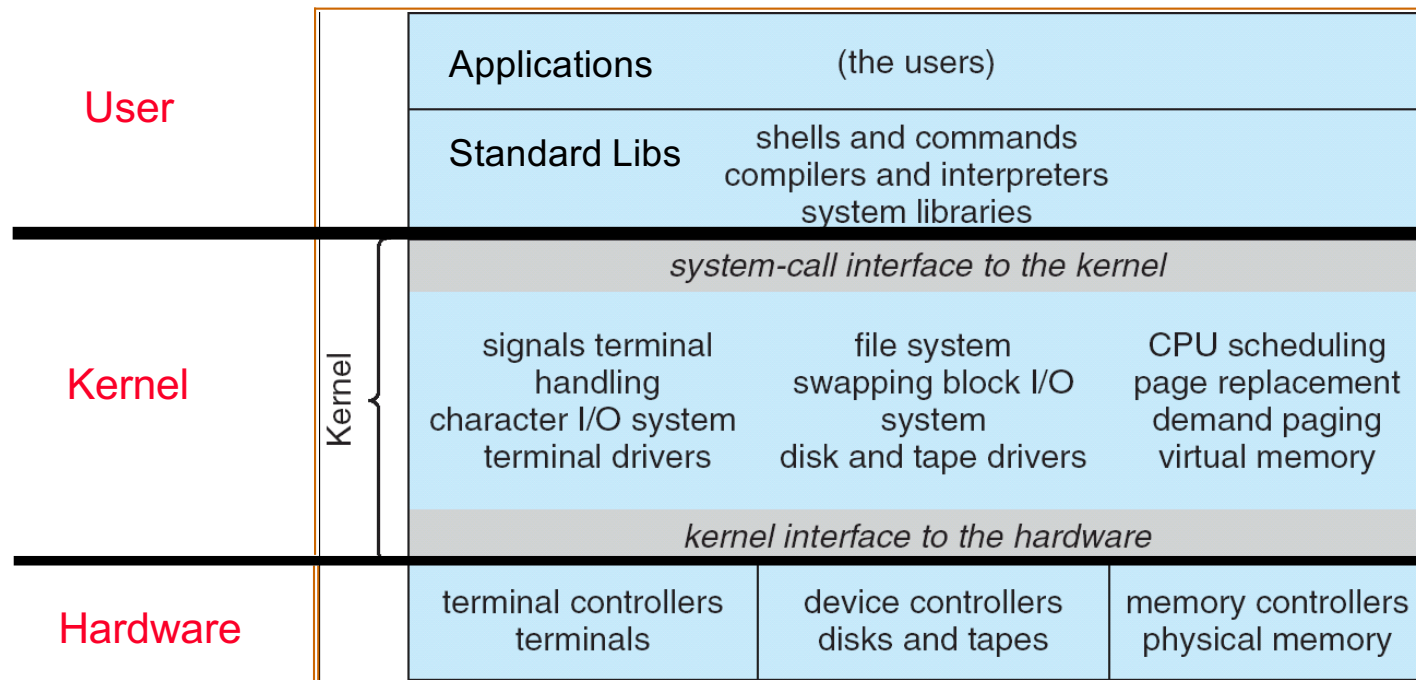
Prof. Yunming XIAO
School of Data Science

Acknowledgments: Ion Stoica and Matei Zaharia, Berkeley CS 162

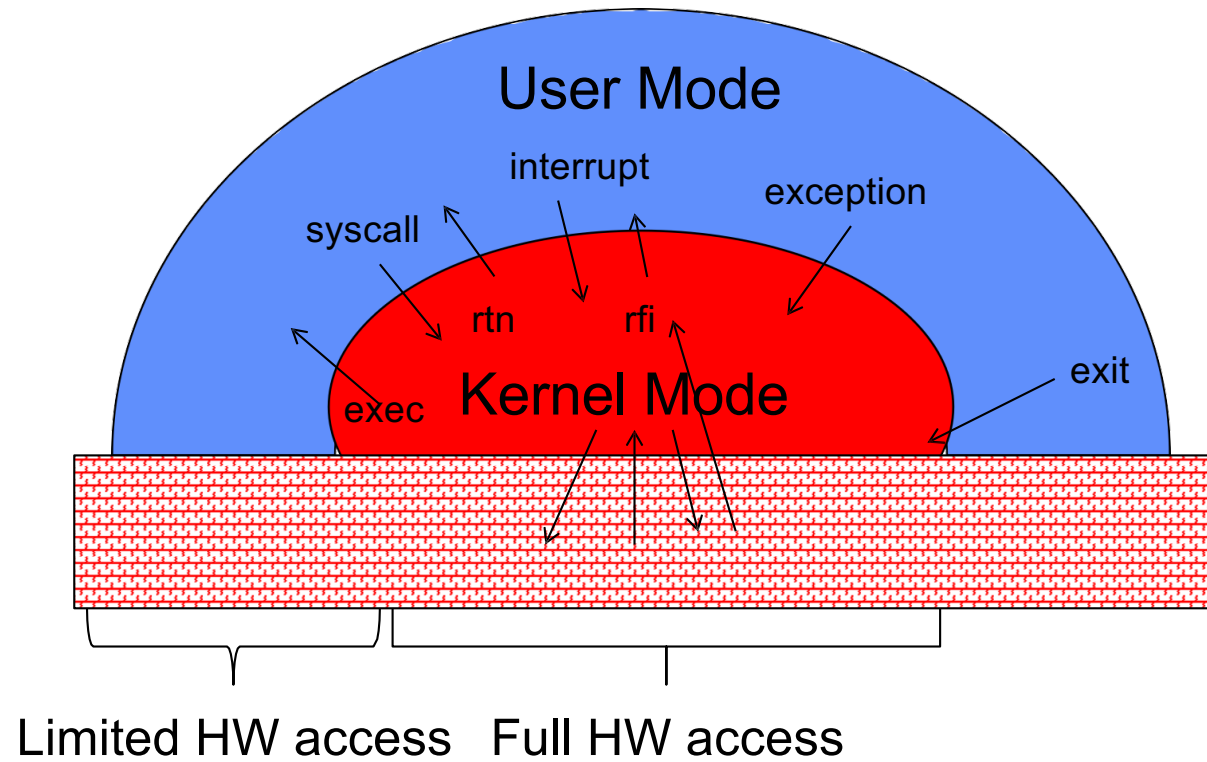
Conclusion: four fundamental OS concepts

- **Process: an instance of a running program**
 - Address Space + one or more Threads
- **Thread: execution context**
 - Fully describes program state
 - Program Counter, Registers, Stack, Execution Flags,
- **Address space (with or w/o translation)**
 - Set of memory addresses accessible to program (for read or write)
 - May be distinct from memory space of the physical machine (in which case programs operate in a virtual address space)
- **Dual mode operation / Protection**
 - Only the “system” has the ability to access certain resources
 - Combined with translation, isolates programs from each other and the OS from programs
 - Key to virtualizing hardware to multiple processes (apps)

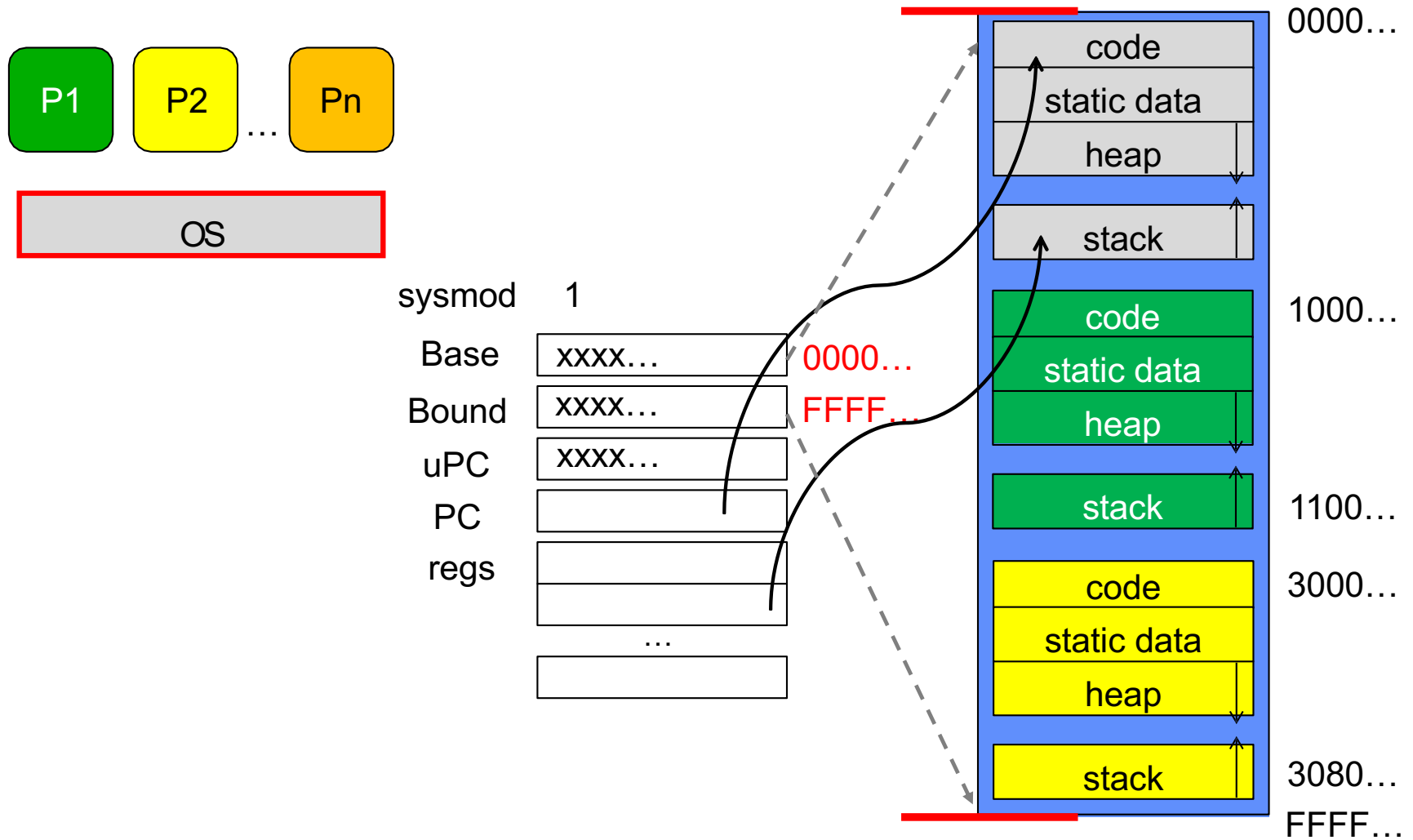
Example: UNIX system structure



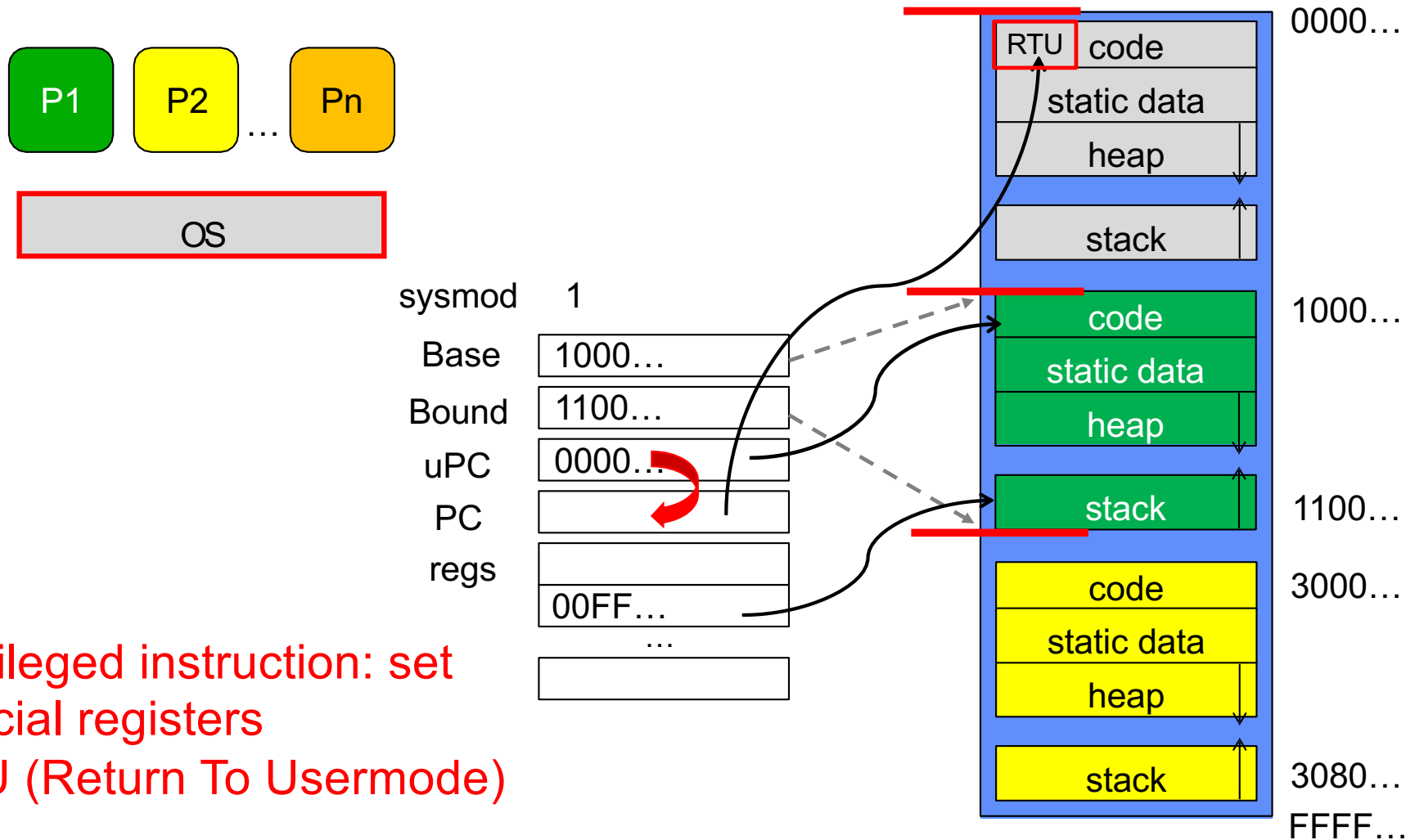
User/Kernel (privileged) mode



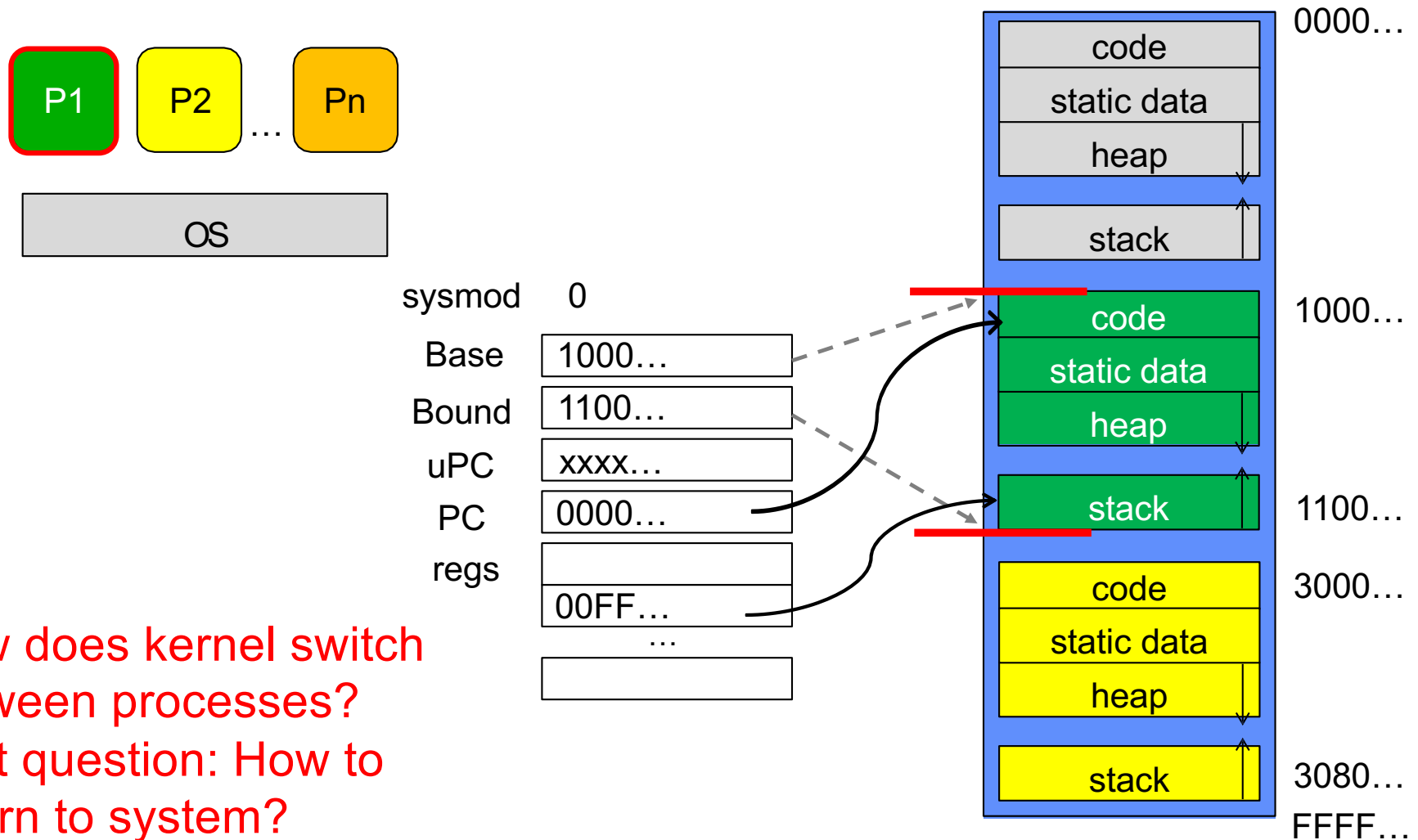
Trying it together: simple B&B: OS load process



Simple B&B: OS gets ready to execute process



Simple B&B: user code running



- How does kernel switch between processes?
- First question: How to return to system?

3 types of user to kernel mode transfer

- **Syscall**
 - Process requests a system service, e.g., exit
 - Like a function call, but “outside” the process
 - Does not have the address of the system function to call
 - Like a Remote Procedure Call (RPC) – for later
 - Marshall the syscall id and args in registers and exec syscall
- **Interrupt**
 - External asynchronous event triggers context switch
 - e. g., Timer, I/O device
 - Independent of user process
- **Trap or Exception**
 - Internal synchronous event in process triggers context switch
 - e.g., Protection violation (segmentation fault), Divide by zero, ...

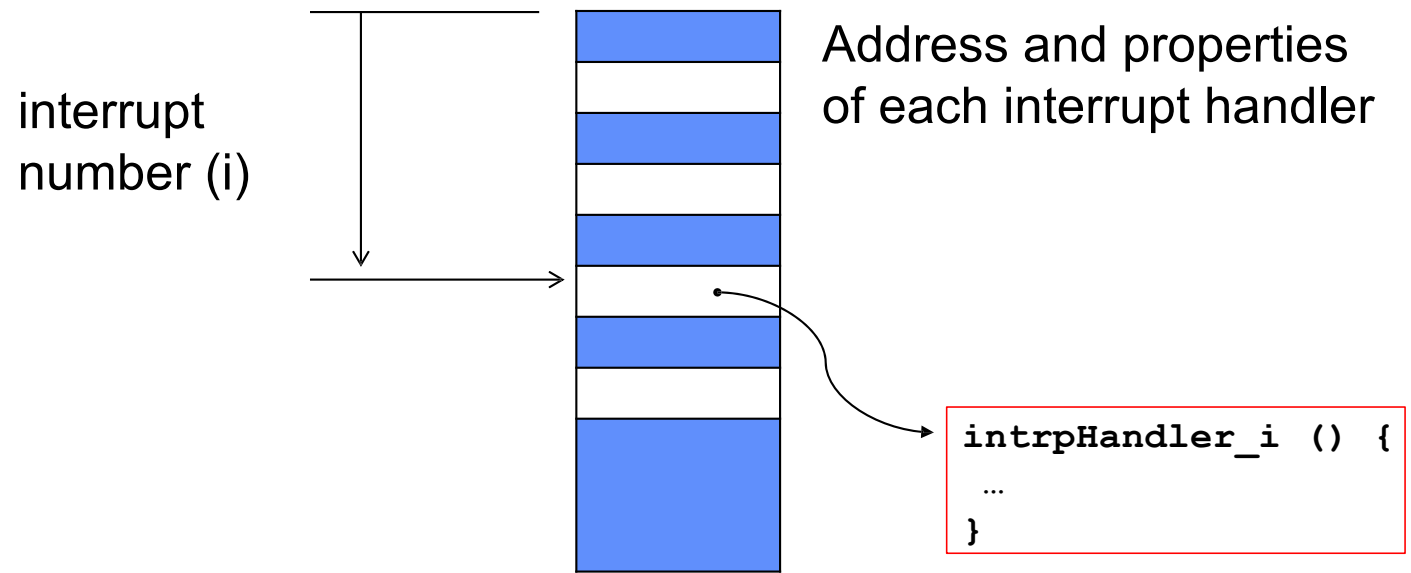
3 types of user to kernel mode transfer

- Syscall
 - Interrupt
 - Trap or Exception
-
- All 3 are an UNPROGRAMMED* CONTROL TRANSFER
 - Where does it go?

*Switching to OS without a direct instruction from the running process

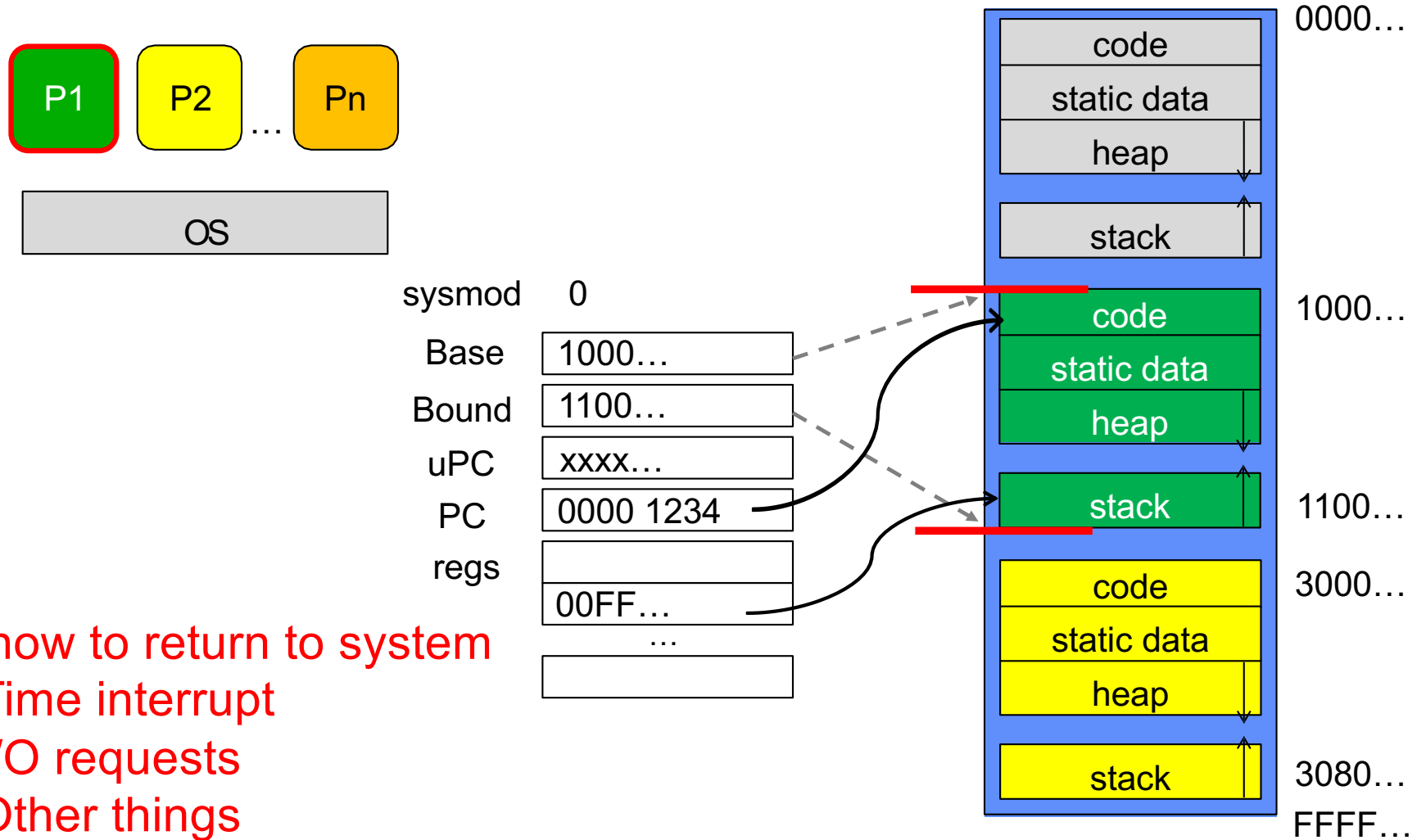
**How do we get the system target address of the
“unprogrammed control transfer”?**

Interrupt vector

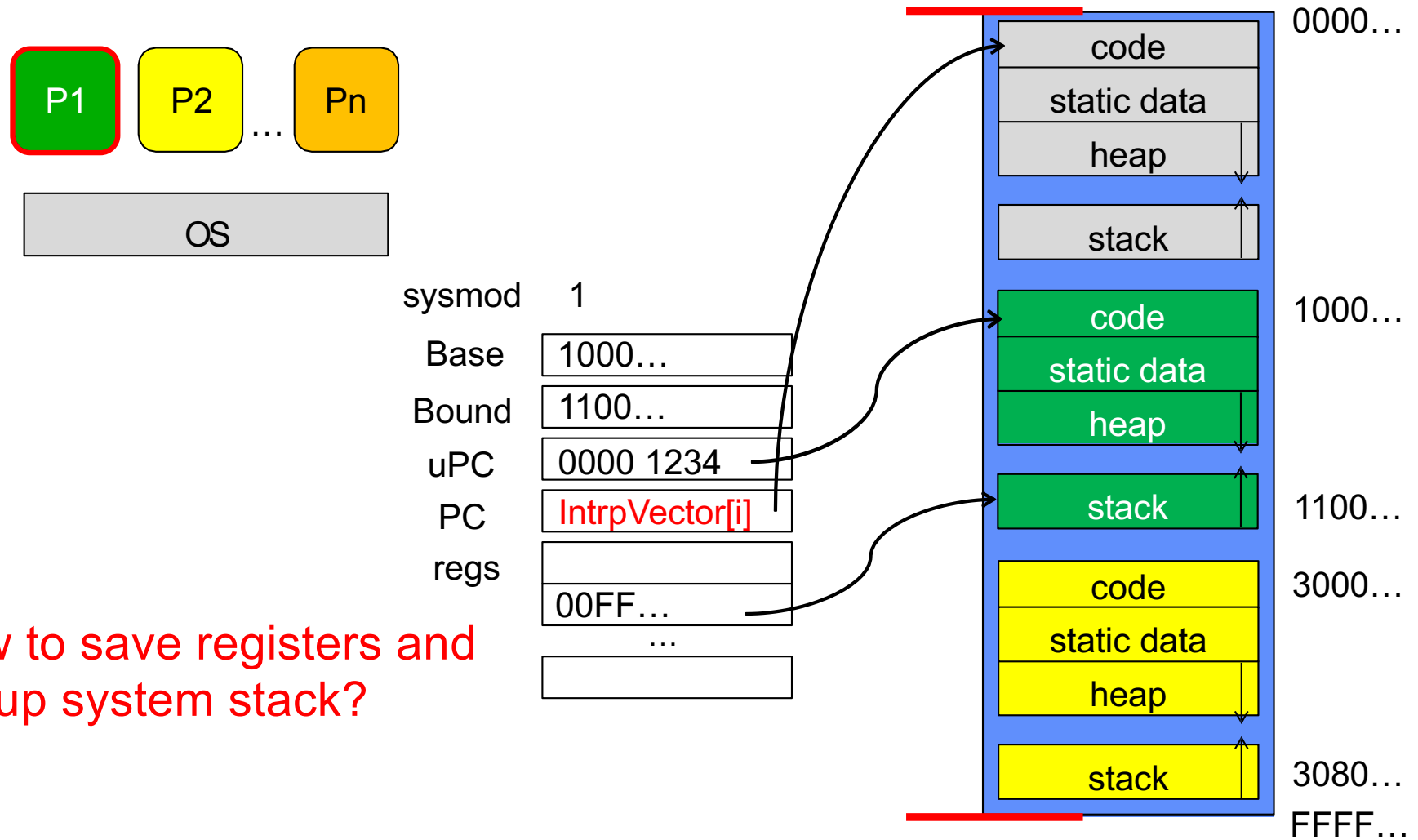


- Where else do you see this dispatch pattern?

Simple B&B: user to kernel

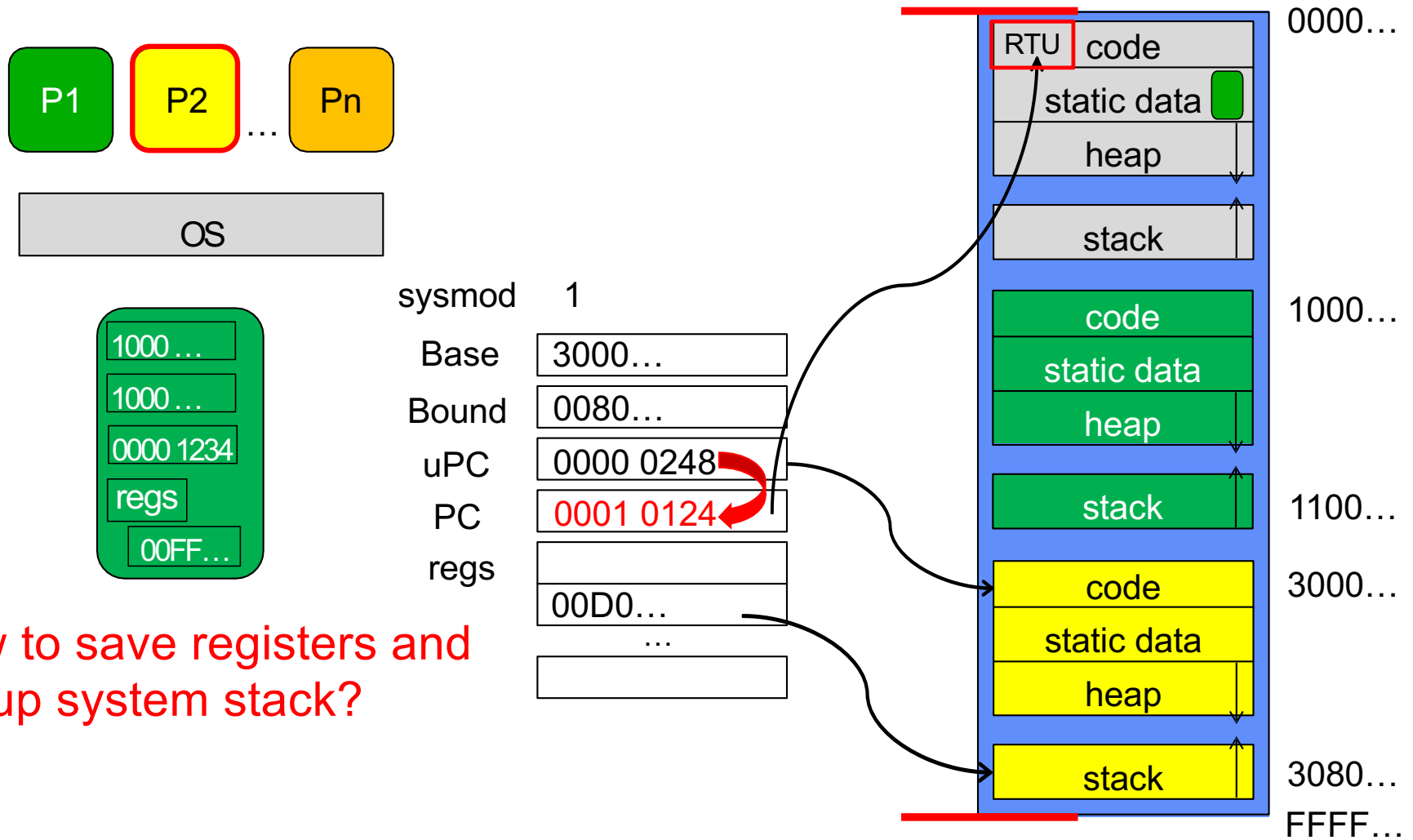


Simple B&B: interrupt



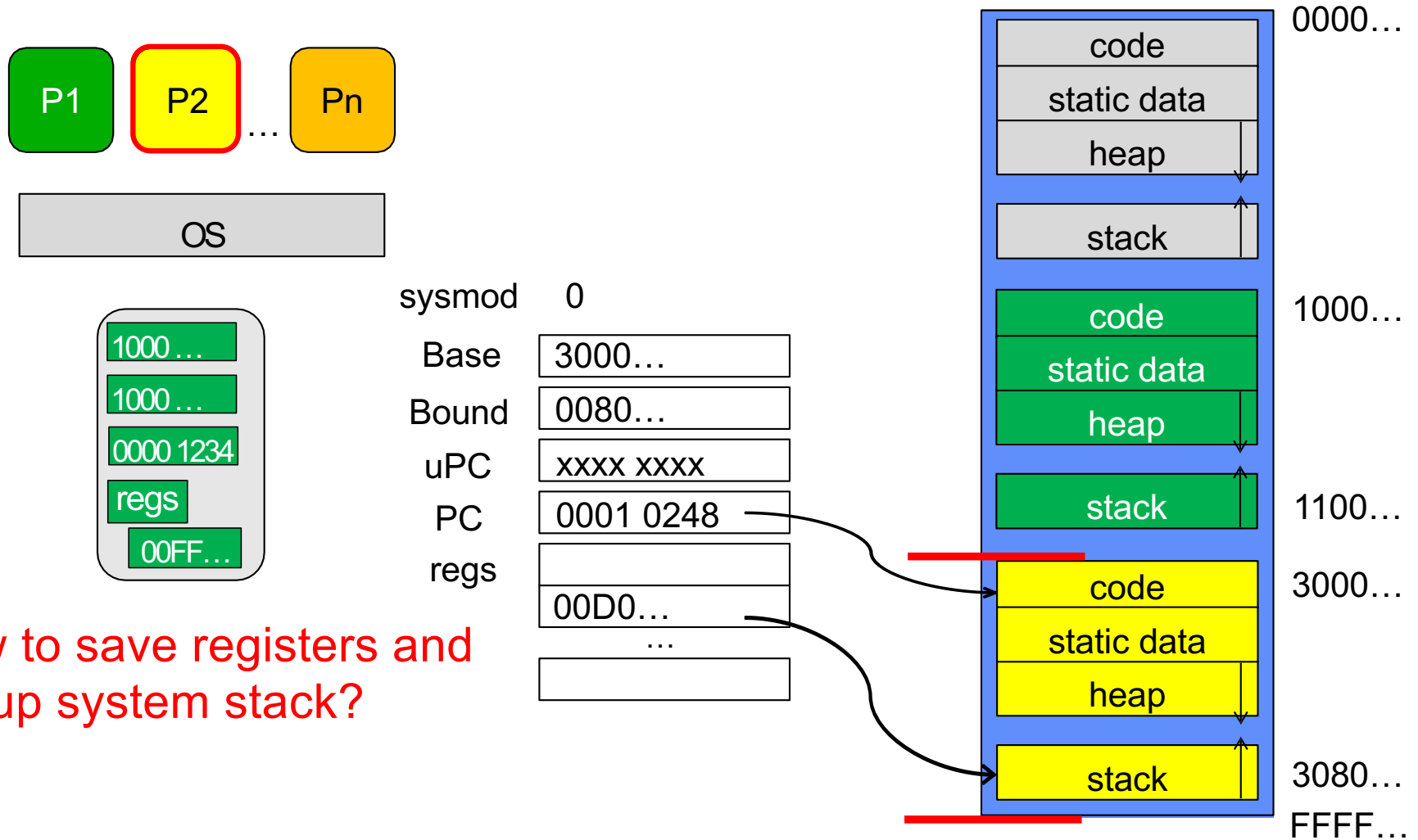
- How to save registers and set up system stack?

Simple B&B: switch user process



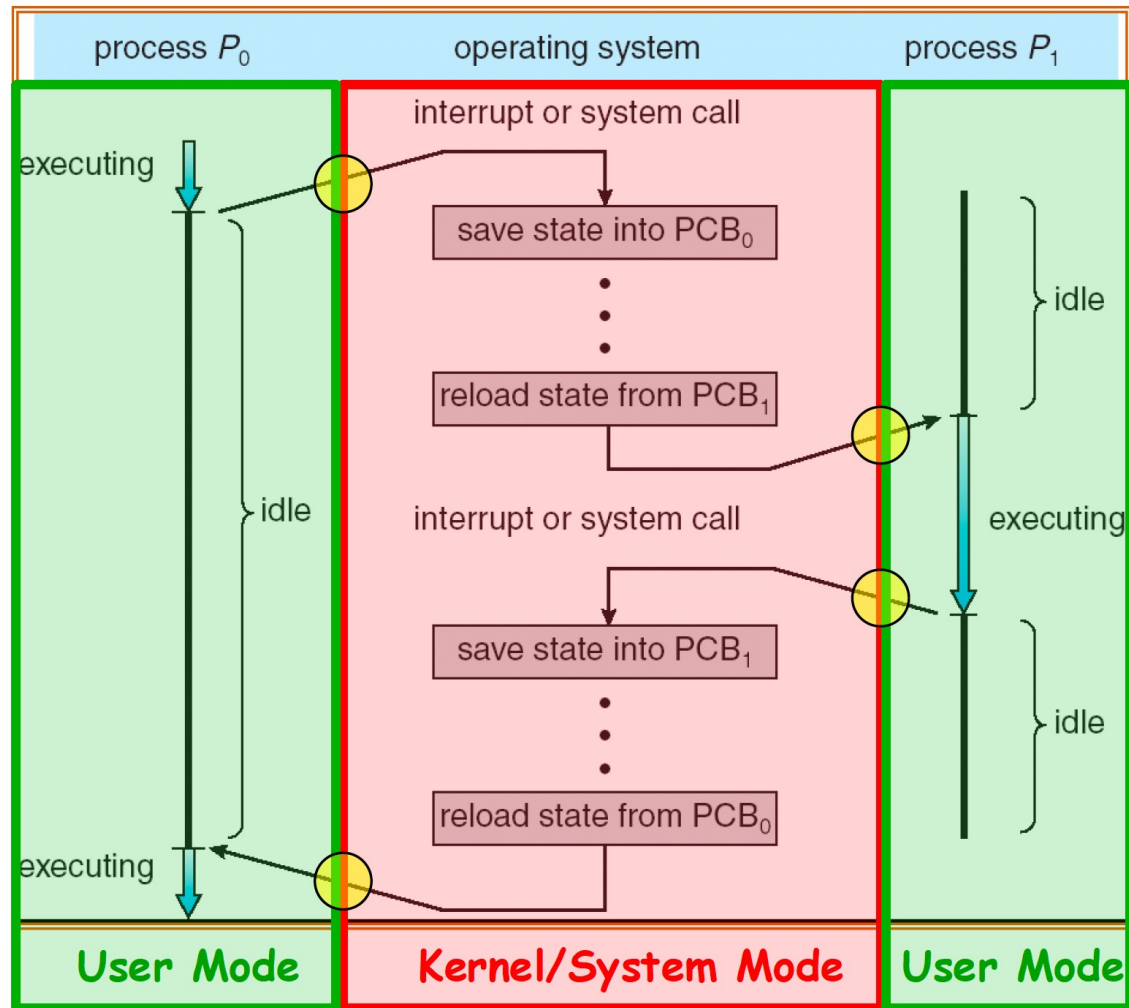
- How to save registers and set up system stack?

Simple B&B: “resume”



- How to save registers and set up system stack?

Another view: CPU switch from process P_0 to process P_1

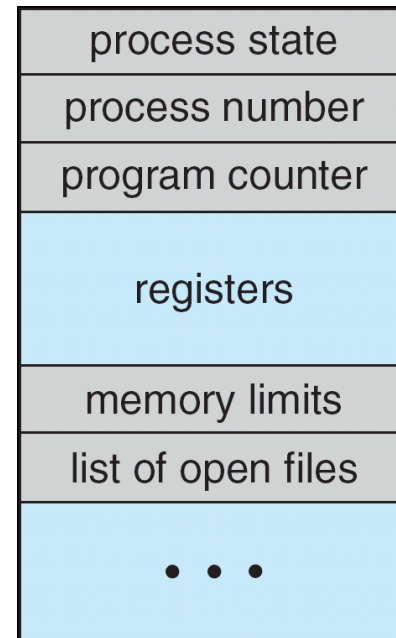


Running many programs?

- We have the basic mechanism to
 - Switch between user processes and the kernel
 - The kernel can switch among user processes
 - Protect OS from user processes and processes from each other
- Questions ?
 - How do we decide which user process to run?
 - How do we represent user processes in the OS?
 - How do we pack up the process and set it aside?
 - How do we get a stack and heap for the kernel?
 - Aren't we wasting a lot of memory?
 - ...

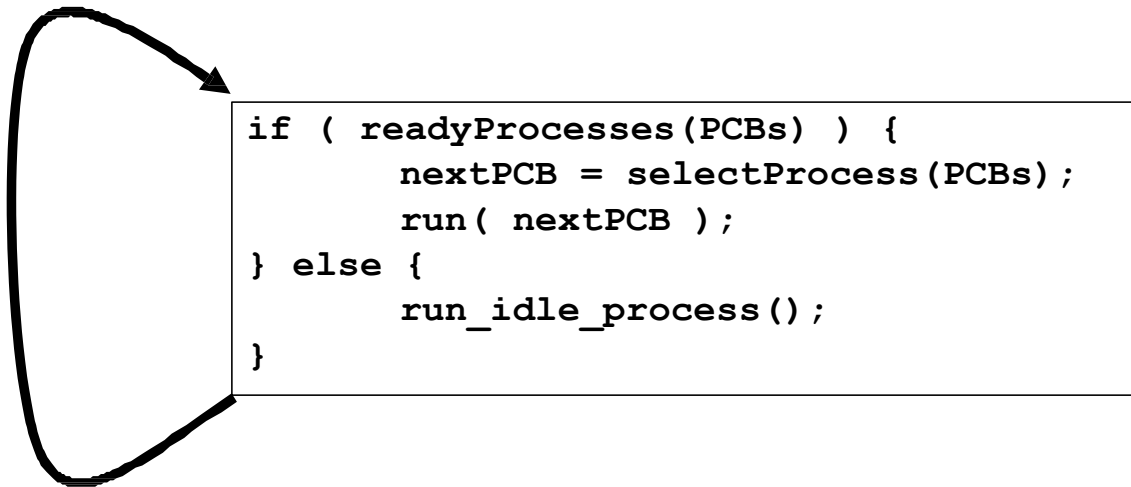
Multiplexing processes: the Process Control Block

- Kernel represents each process with a process control block (PCB)
 - Status (running, ready, blocked, ...)
 - Register state (when not ready)
 - Process ID (PID), User, Executable, Priority, ...
 - Execution time, ...
 - Memory space, translation, ...
- Kernel Scheduler maintains a data structure containing the PCBs
 - Give out CPU to different processes
 - This is a Policy Decision
- Give out non-CPU resources
 - Memory/IO
 - Another policy decision



Process Control Block

Scheduler



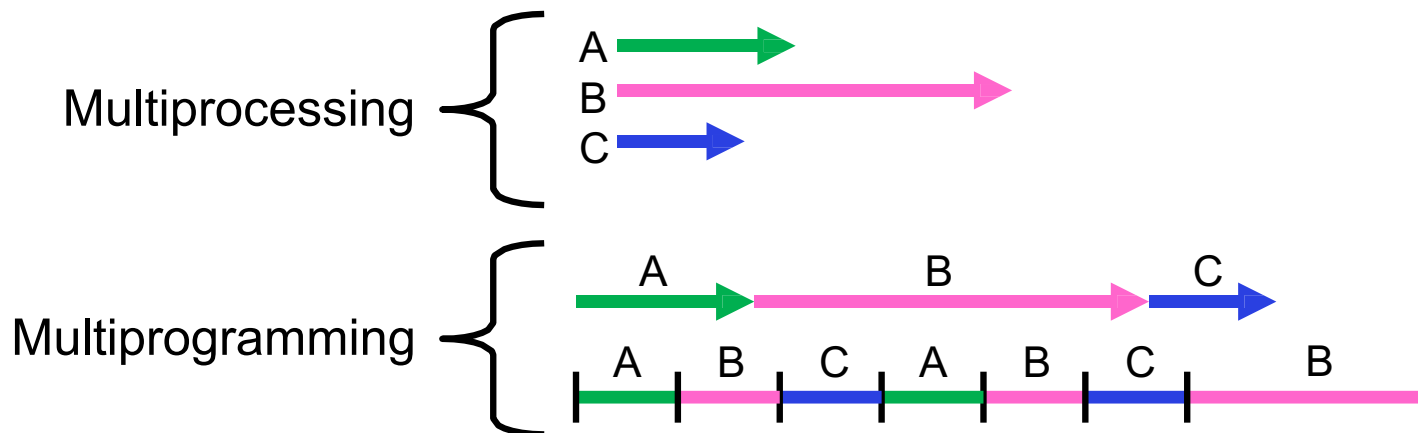
- Scheduling: mechanism for deciding which processes/threads receive hardware CPU time, when, and for how long
- Lots of different scheduling policies provide ...
 - Fairness
 - Realtime guarantees
 - Latency optimization
 - ...

Going back to threads

- Definition from before: *a single unique execution context*
 - Describes its representation
- It provides the abstraction of: *a single execution sequence that represents a separately schedulable task*
 - Also a valid definition!
- Threads are a mechanism for *concurrency* (overlapping execution)
 - However, they can also run in *parallel* (simultaneous execution)
- Protection is an orthogonal concept
 - A protection domain (i.e., process) can contain one thread or many

Multiprocessing vs. multiprogramming

- Some definitions:
 - Multiprocessing: multiple CPUs (cores)
 - Multiprogramming: multiple jobs/processes
 - Multithreading: multiple threads/processes
- What does it mean to run two threads concurrently?
 - Scheduler is free to run threads in any order and interleaving
 - Thread may run to completion or time-slice in big chunks or small chunks



Concurrency is not parallelism

- Concurrency is about handling multiple things at once
- Parallelism is about doing multiple things simultaneously
- Example: Two threads on a single-core system...
 - ... execute concurrently ...
 - ... but not in parallel
- Each thread handles or manages a separate thing or task...
- But those tasks are not necessarily executing simultaneously!

Silly example for threads

- Imagine the following program:

```
main() {  
    ComputePI("pi.txt");  
    PrintClassList("classlist.txt");  
}
```

- What is the behavior here?
- Program would never print out class list
- Why? ComputePI would never finish

Adding threads

- Version of program with threads (loose syntax):

```
main() {  
    create_thread(ComputePI, "pi.txt");  
    create_thread(PrintClassList, "classlist.txt");  
}
```

→ T1*

→ T2

- `create_thread`: spawns a new thread running the given procedure
 - Should behave as if another CPU is running the given procedure
- Now, you would actually see the class list



More practical motivation: compute/I/O overlap

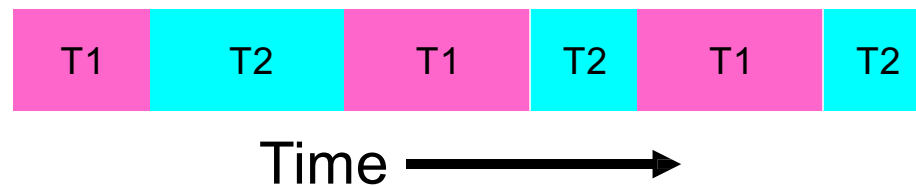
- Back to Jeff Dean's "Numbers Everyone Should Know"

L1 cache reference	0.5 ns
Branch mispredict	5 ns
L2 cache reference	7 ns
Mutex lock/unlock	25 ns
Main memory reference	100 ns
Compress 1K bytes with Zip	3,000 ns
Send 2K bytes over 1 Gbps network	20,000 ns
Read 1 MB sequentially from memory	250,000 ns
Round trip within same datacenter	500,000 ns
Disk seek	10,000,000 ns
Read 1 MB sequentially from disk	20,000,000 ns
Send packet CA->Netherlands->CA	150,000,000 ns

- Handle I/O in a separate thread, avoiding blocking other progress

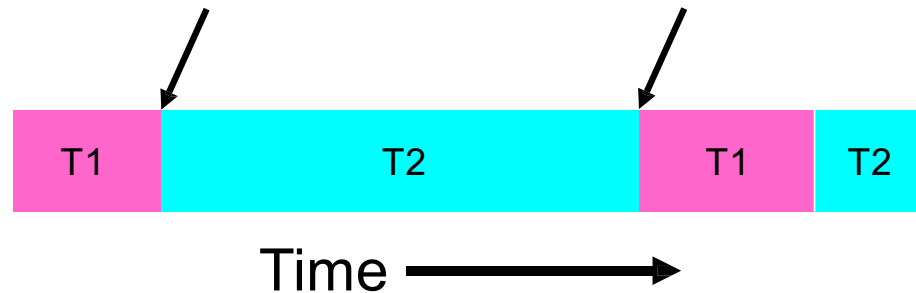
Threads mask I/O latency

- If no thread performs I/O:



- If thread 1 performs blocking I/O operation:

T1 starts I/O operation T1 I/O completes



A better example of threads

```
main() {  
    create_thread(ReadLargeFile, "pi.txt");  
    create_thread(RenderUserInterface);  
}
```

- What is the behavior here?
 - Still respond to user input
 - While reading file in the background

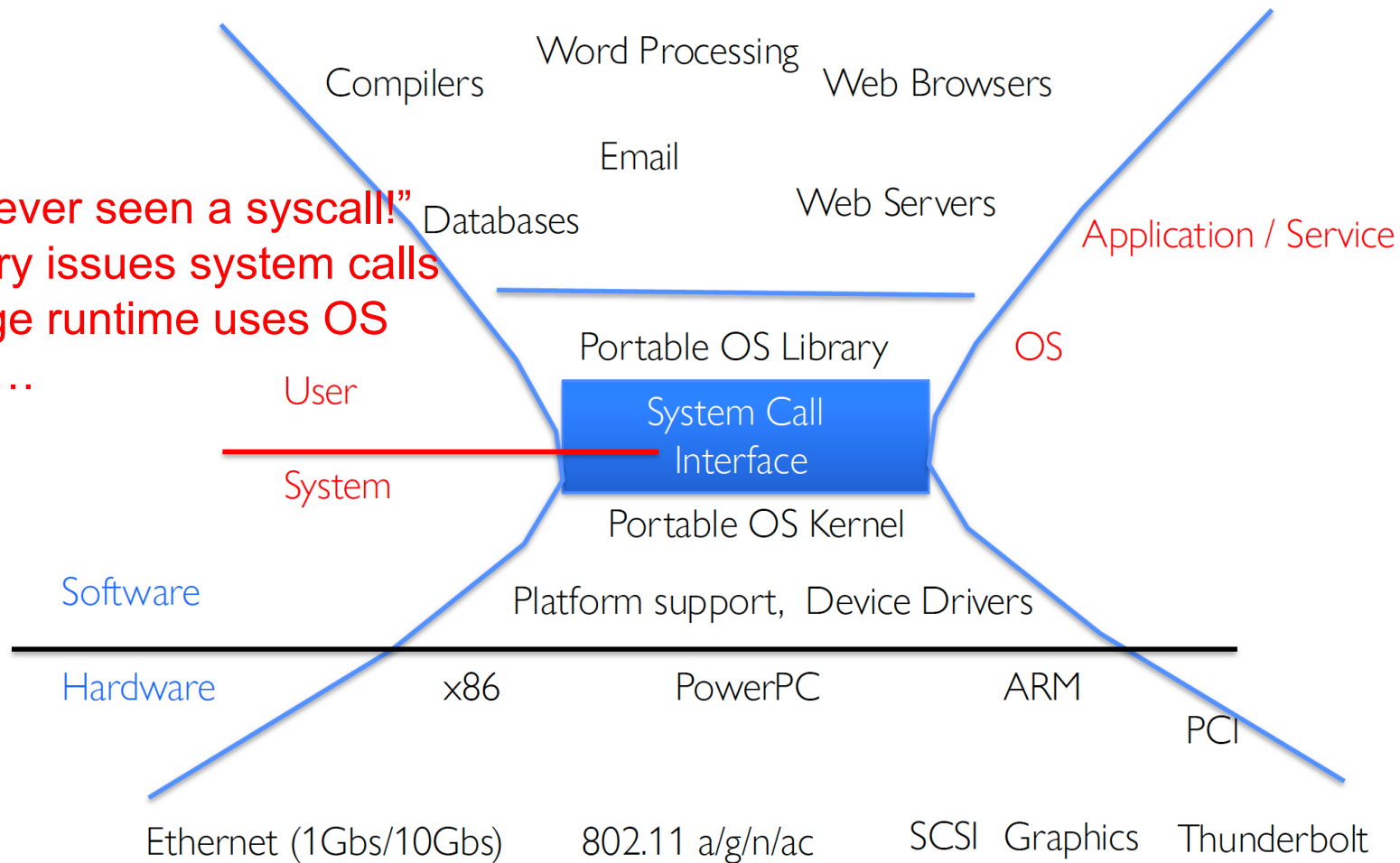
Multithreaded programs

- You know how to compile a C program and run the executable
 - This creates a process that is executing that program
- Initially, this new process has one thread in its own address space
 - With code, global variables, etc. as specified in the executable
- Q: How can we make a multithreaded process?
- A: Once the process starts, it issues system calls to create new threads
 - These new threads are part of the process: they share its address space

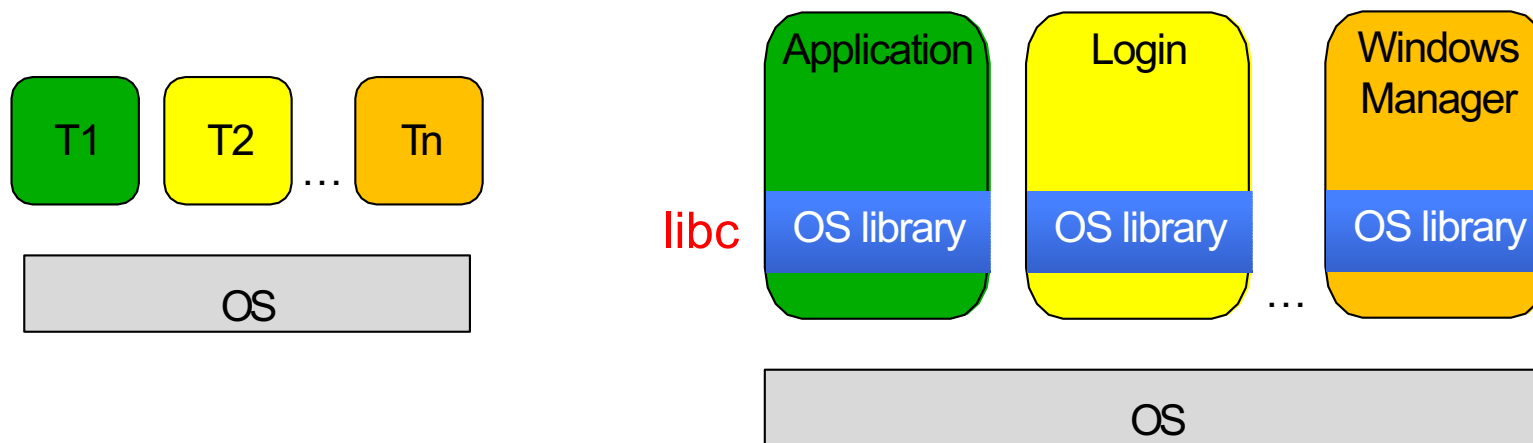
System calls (“syscalls”)

“But I’ve never seen a syscall!”

- OS library issues system calls
- Language runtime uses OS libraries...



OS library issues syscalls



OS library API for threads: *pthread*

(Here, the “p” is for “POSIX” which is a part of standardized API)

```
int pthread_create(pthread_t *thread, const pthread_attr_t *attr,  
                  void *(*start_routine)(void*), void *arg);
```

- thread is created executing start_routine with arg as its sole argument.
- return is implicit call to pthread_exit
- (attr contains info like stack size, scheduling policy, etc)

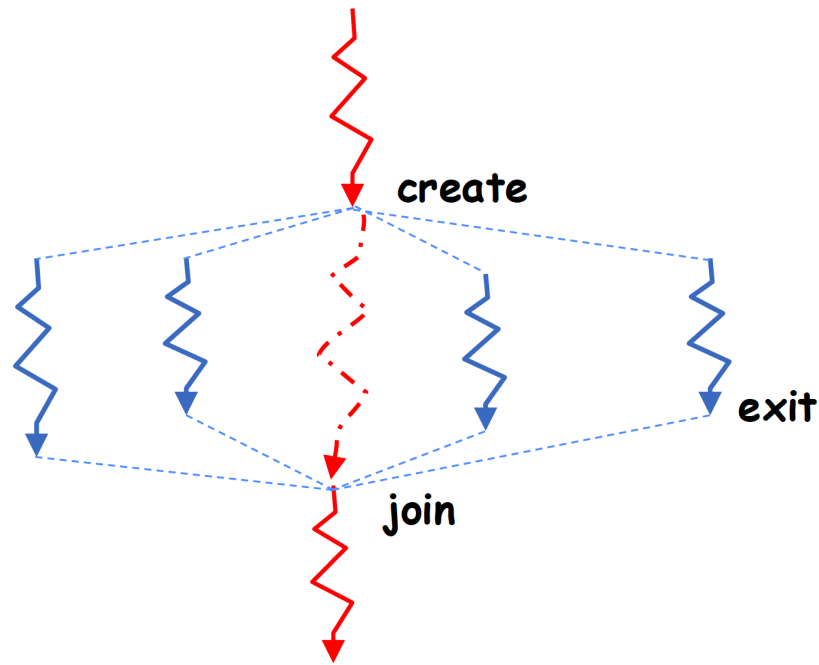
```
void pthread_exit(void *value_ptr);
```

- terminates the thread and makes value_ptr available to any successful join

```
int pthread_join(pthread_t thread, void **value_ptr);
```

- suspends execution of the calling thread until the target thread terminates.
- On return with a non-NULL value_ptr the value passed to pthread_exit() by the terminating thread is made available in the location referenced by value_ptr.

New idea: fork-join pattern



- Main thread *creates* (forks) collection of sub-threads passing them args to work on...
- ... and then *joins* with them, collecting results.

Pthread examples

- How many threads are in this program?
- What function does each thread run?
- One possible result:

```
((base) CullerMac19:code04 culler$ ./pthread 4
Main stack: 7ffee2c6b6b8, common: 10cf95048 (162)
Thread #1 stack: 70000d83bef8 common: 10cf95048 (162)
Thread #3 stack: 70000d941ef8 common: 10cf95048 (164)
Thread #2 stack: 70000d8beef8 common: 10cf95048 (165)
Thread #0 stack: 70000d7b8ef8 common: 10cf95048 (163)
```

- Does the main thread join with the threads in the same order that they were created?
 - Yes: Loop calls Join in thread order
- Do the threads exit in the same order they were created?
 - No: Depends on scheduling order!
- Would the result change if run again?
 - Yes: Depends on scheduling order!
- Is this code safe/correct???
 - No – threads share a variable that is used
- without locking and there is a race condition!

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <string.h>

int common = 162;

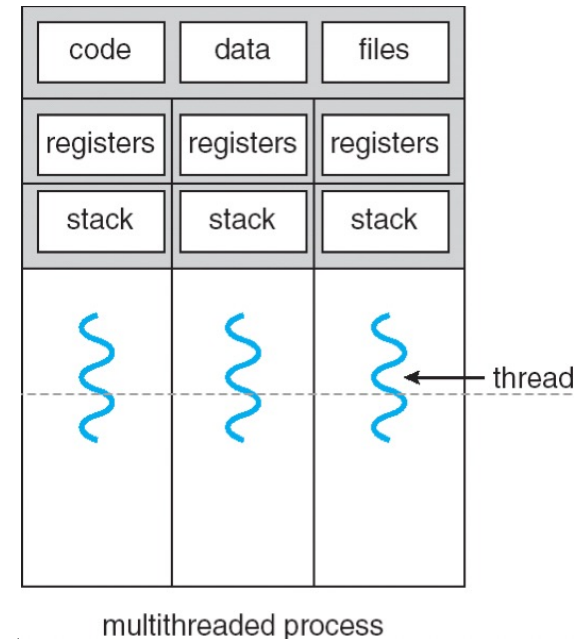
void *threadfun(void *threadid)
{
    long tid = (long)threadid;
    printf("Thread #%lx stack: %lx common: %lx (%d)\n", tid,
        (unsigned long) &tid, (unsigned long) &common, common++);
    pthread_exit(NULL);
}

int main (int argc, char *argv[])
{
    long t;
    int nthreads = 2;
    if (argc > 1) {
        nthreads = atoi(argv[1]);
    }
    pthread_t *threads = malloc(nthreads*sizeof(pthread_t));
    printf("Main stack: %lx, common: %lx (%d)\n",
        (unsigned long) &t, (unsigned long) &common, common);
    for(t=0; t<nthreads; t++){
        int rc = pthread_create(&threads[t], NULL, threadfun, (void *)t);
        if (rc){
            printf("ERROR: return code from pthread_create() is %d\n", rc);
            exit(-1);
        }
    }

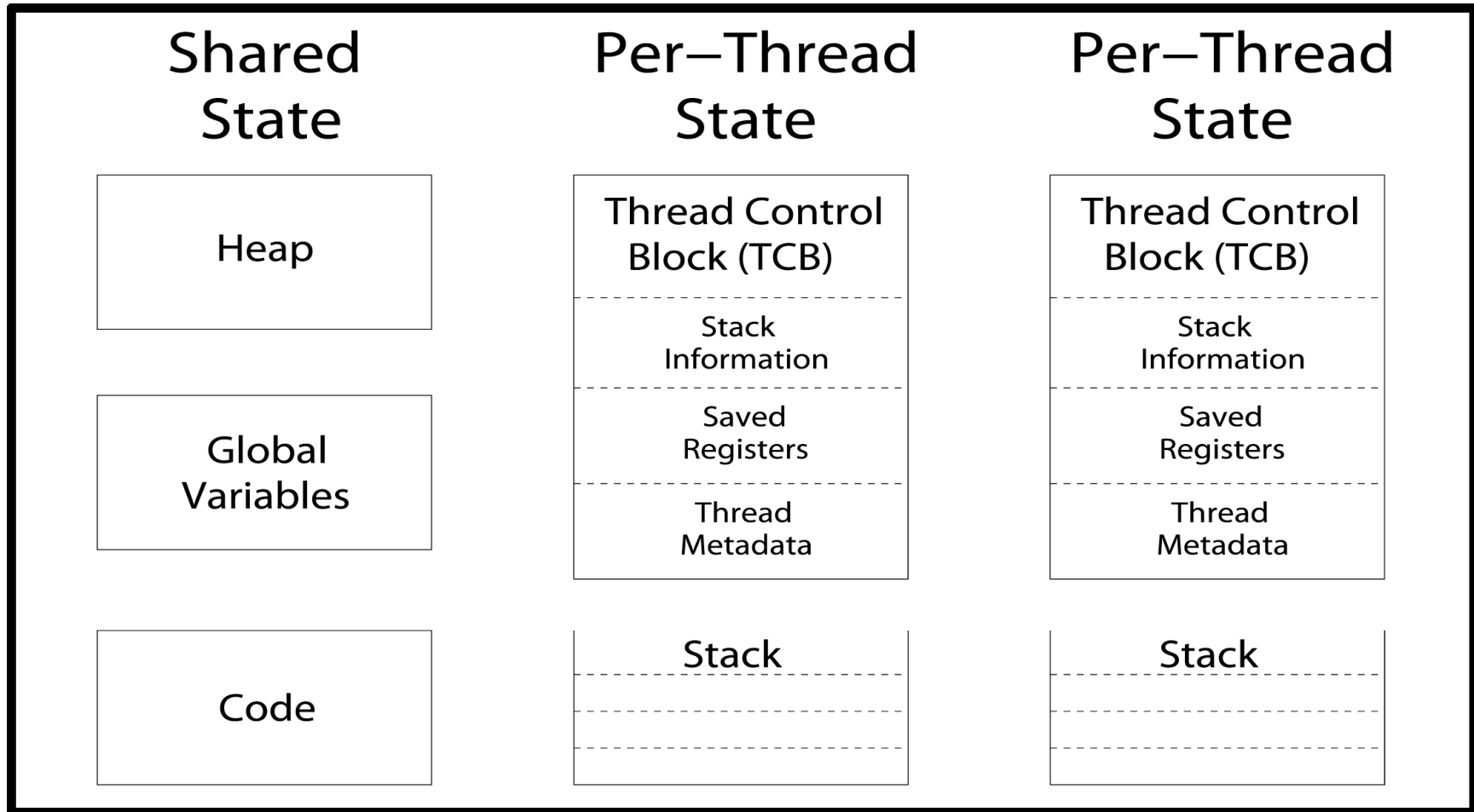
    for(t=0; t<nthreads; t++){
        pthread_join(threads[t], NULL);
    }
    pthread_exit(NULL);
} /* last thing in the main thread */
```

Thread state

- State shared by all threads in process/address space
 - Content of memory (global variables, heap)
 - I/O state (file descriptors, network connections, etc)
- State “private” to each thread
 - Kept in **TCB (Thread Control Block)**
 - CPU registers (including, program counter)
 - Execution stack
- Execution Stack
 - Parameters, temporary variables
 - Return PCs are kept while called procedures are executing



Shared vs. pre-thread state



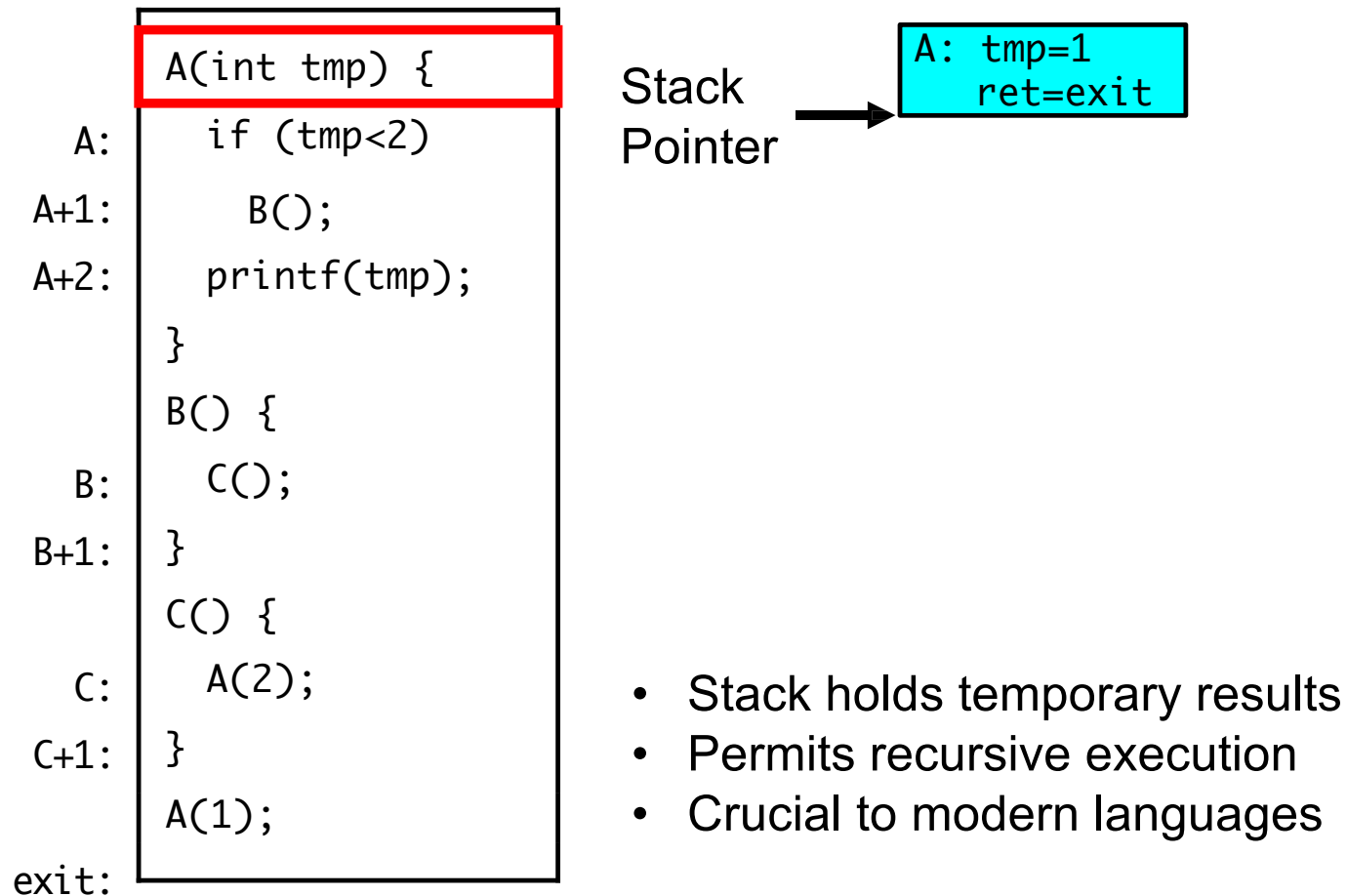
Process State

Execution stack example

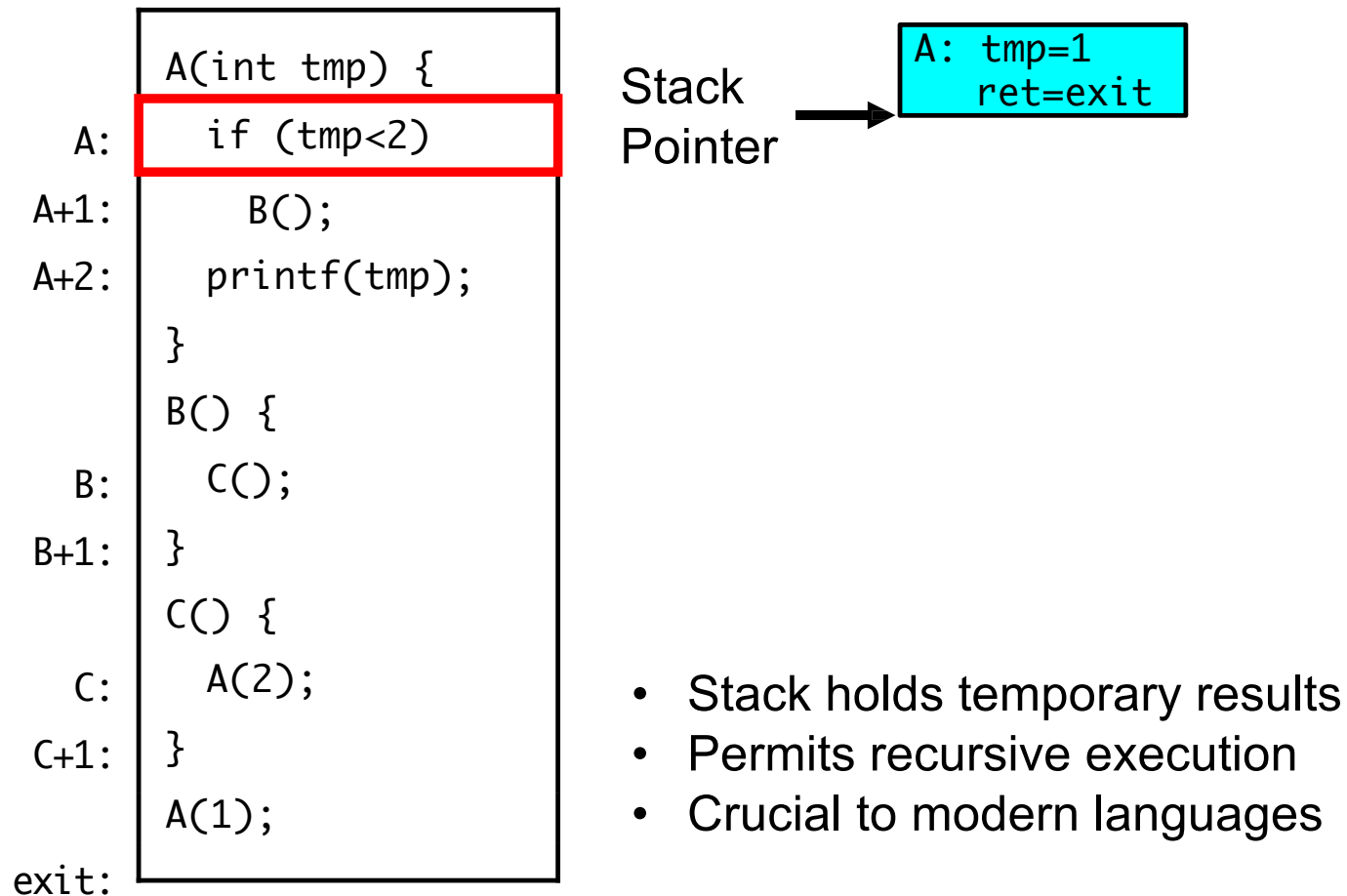
```
    A(int tmp) {  
A:    if (tmp<2)  
A+1:    B();  
A+2:    printf(tmp);  
    }  
    B() {  
B:    C();  
B+1: }  
    C() {  
C:    A(2);  
C+1: }  
    A(1);  
exit:
```

- Stack holds temporary results
- Permits recursive execution
- Crucial to modern languages

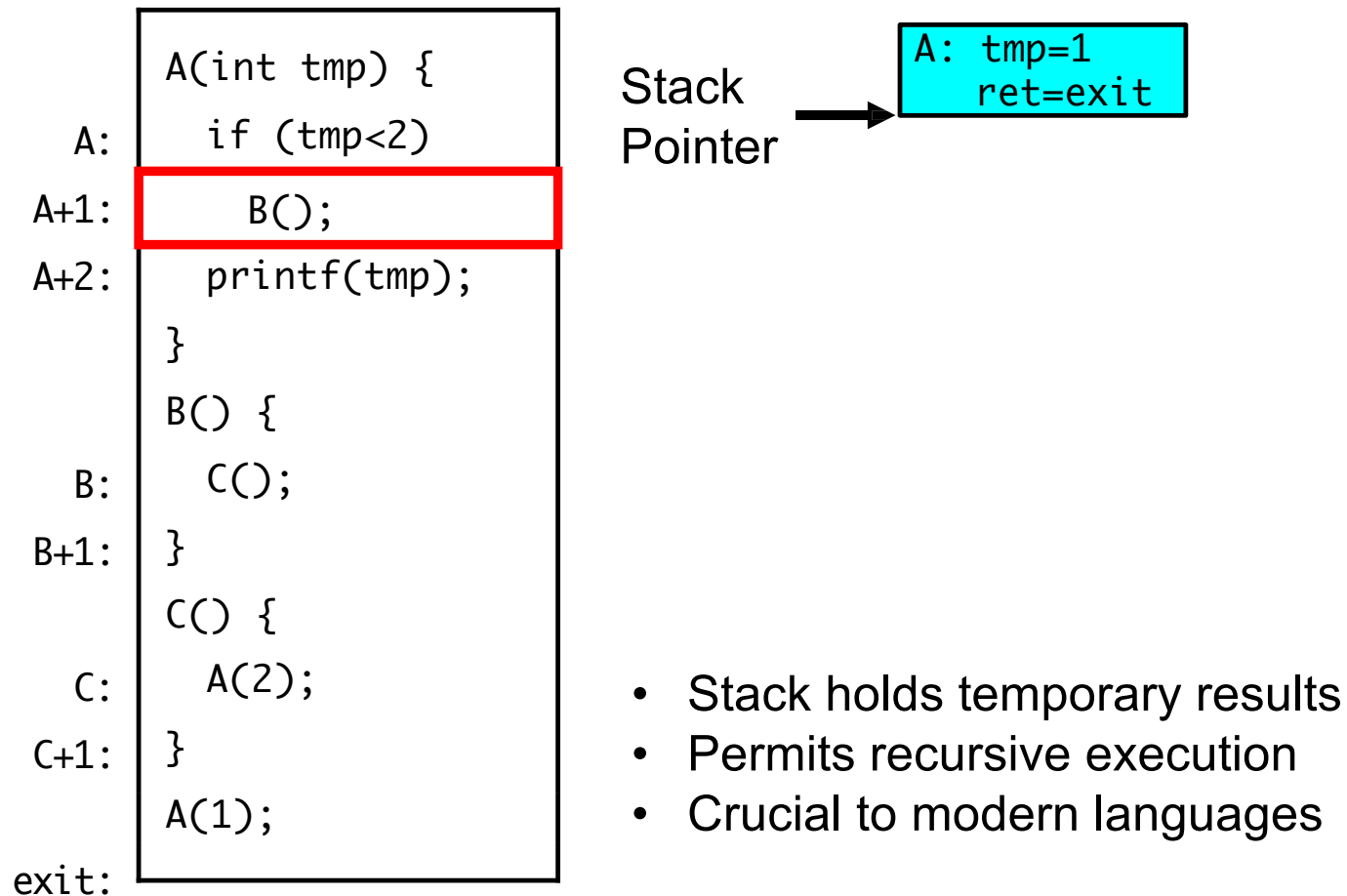
Execution stack example



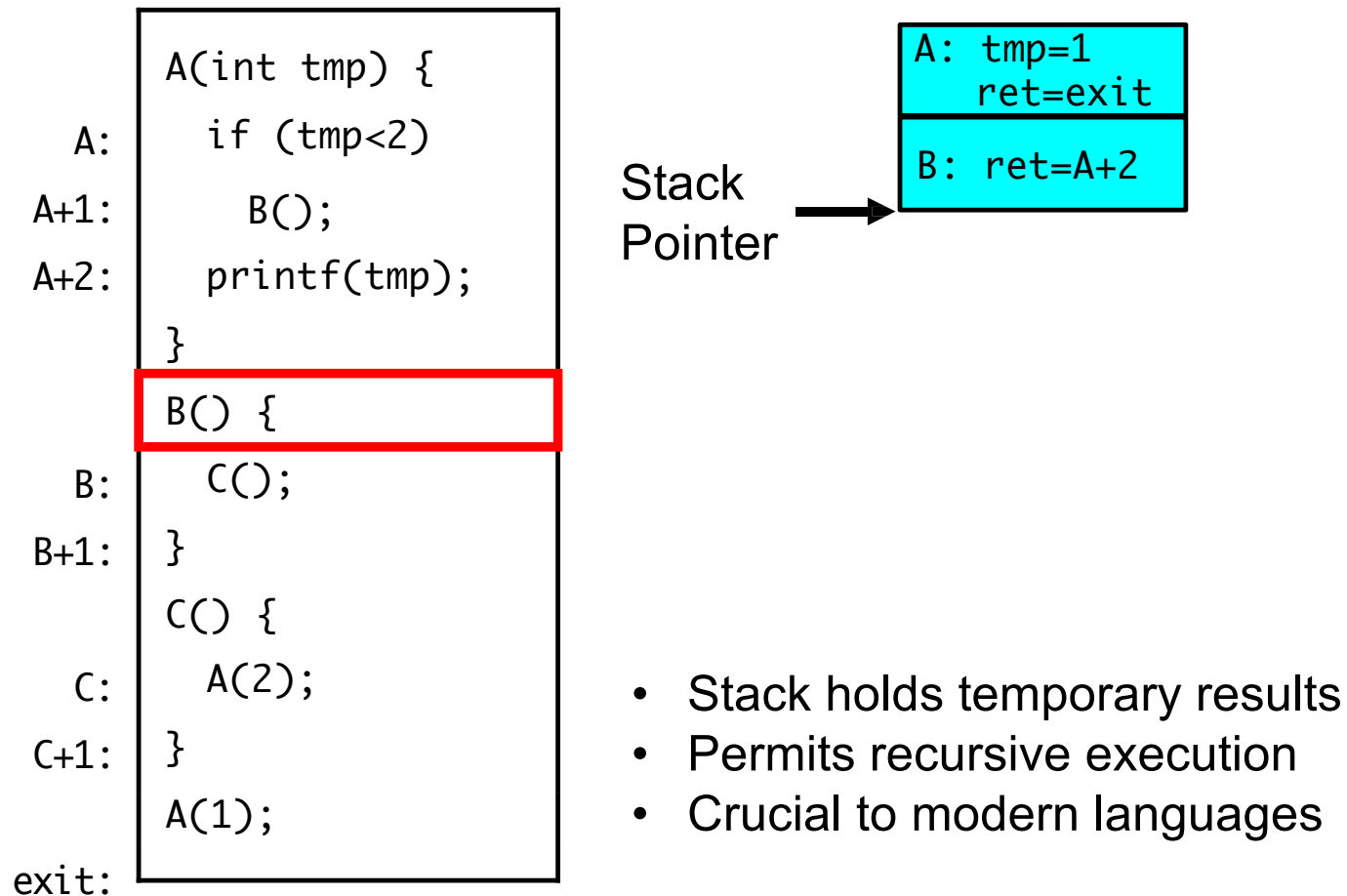
Execution stack example



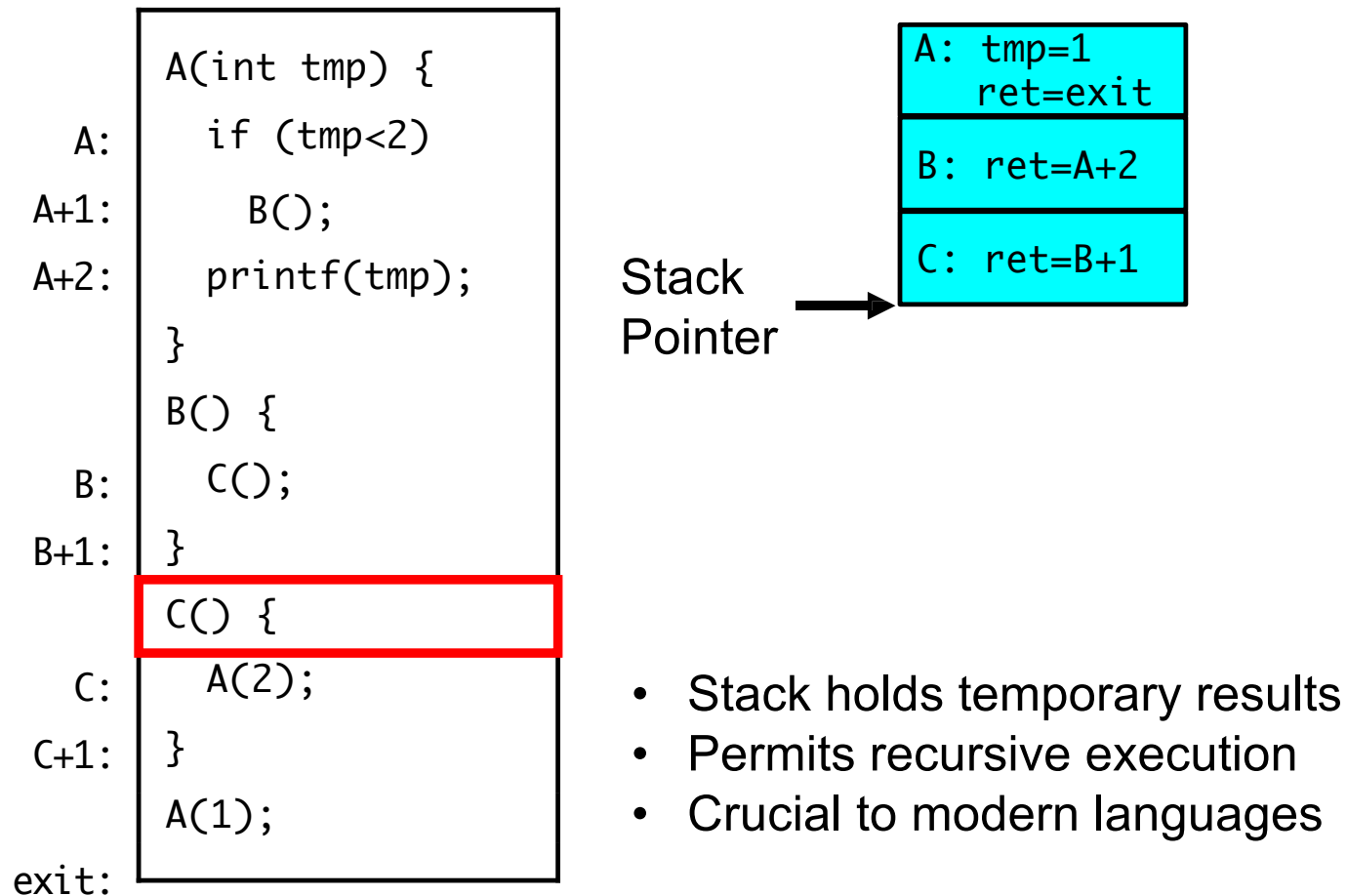
Execution stack example



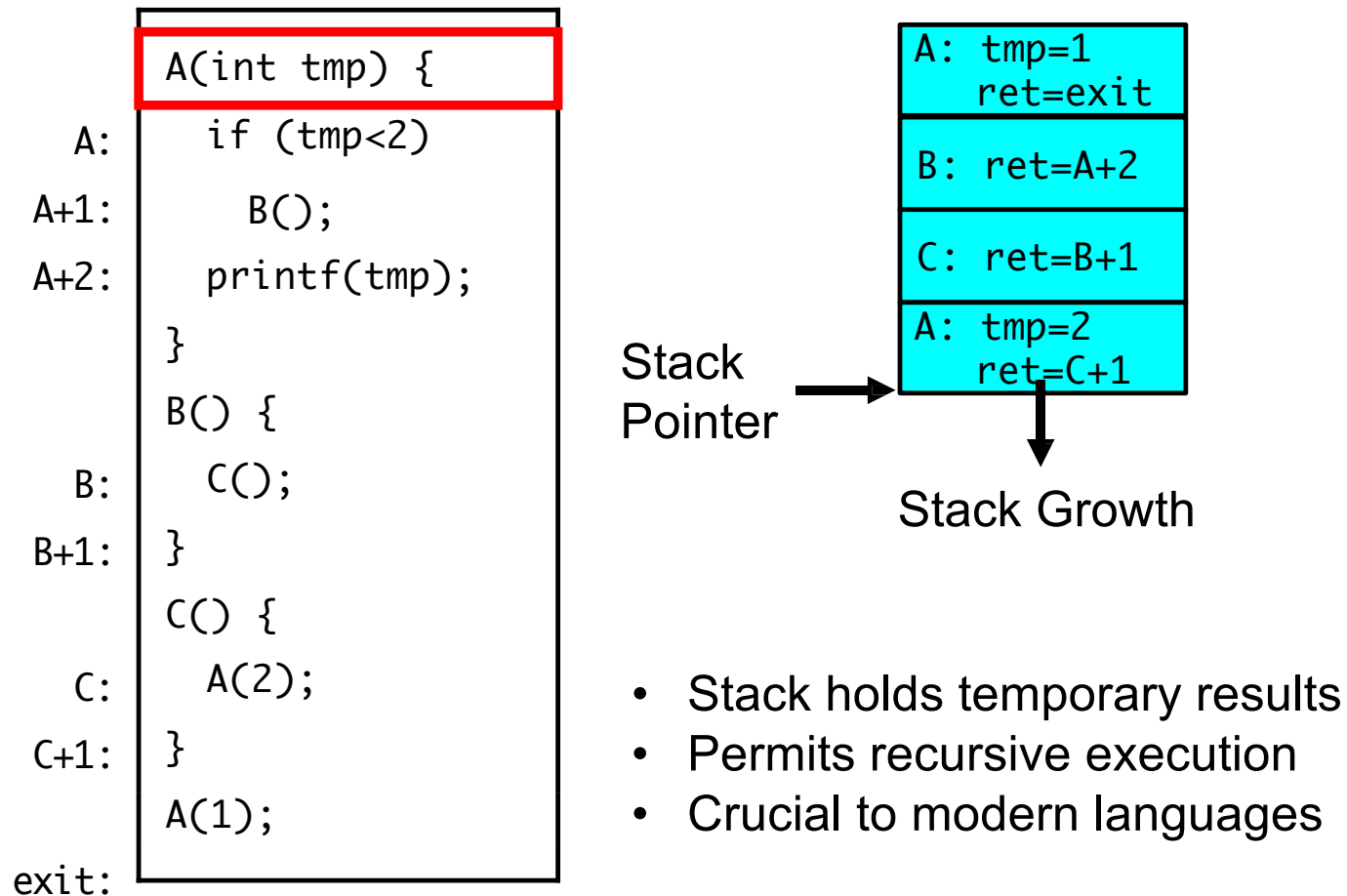
Execution stack example



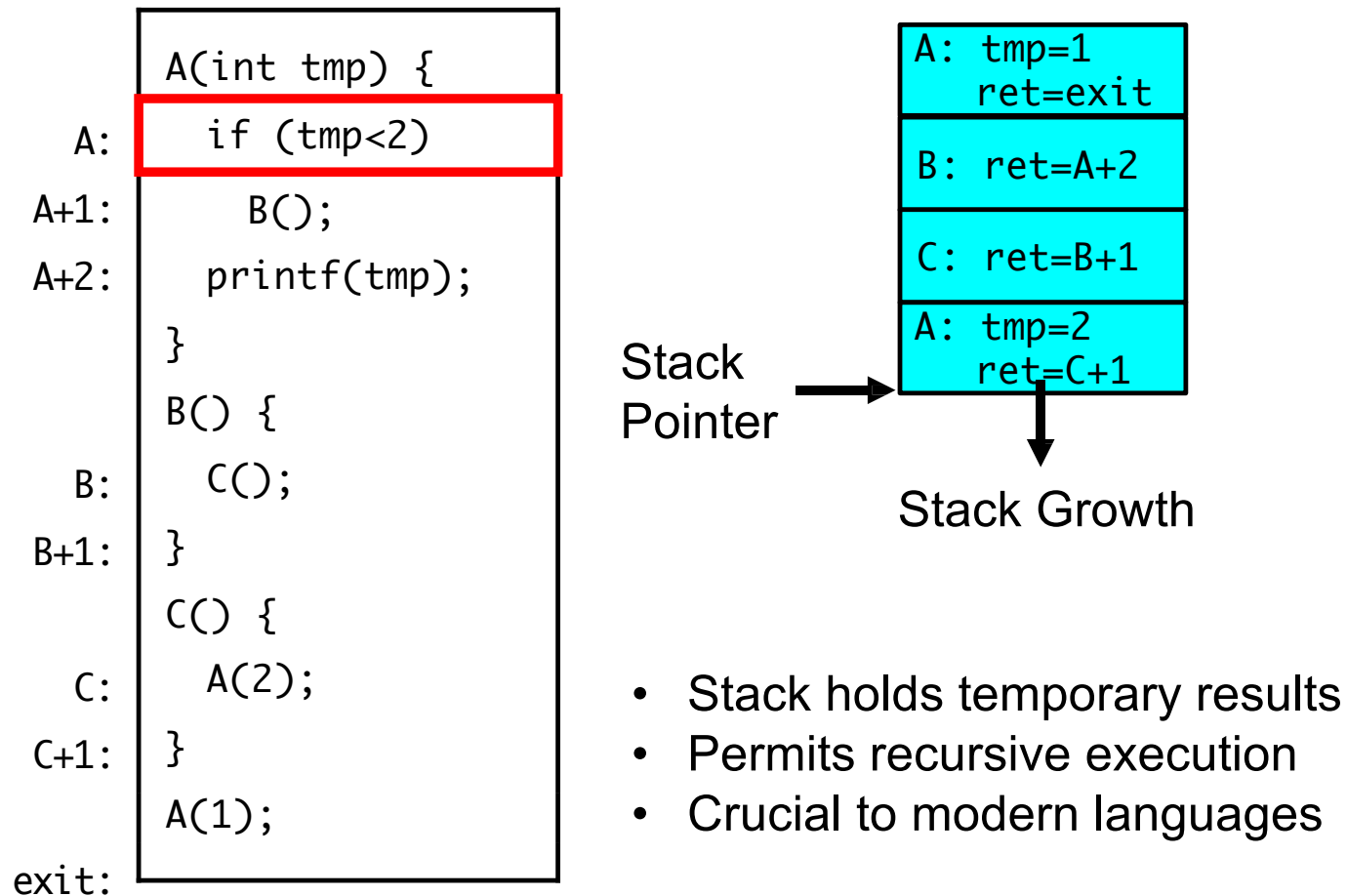
Execution stack example



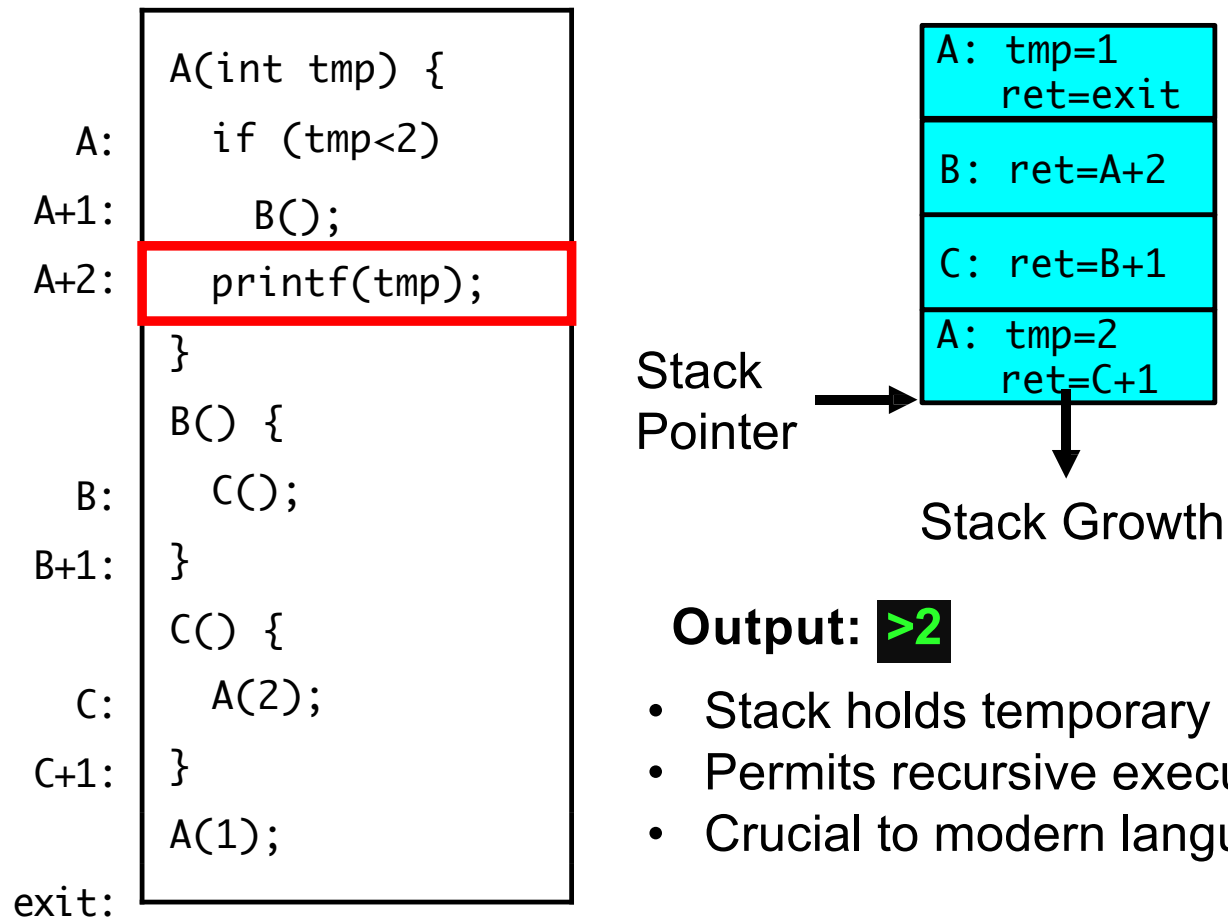
Execution stack example



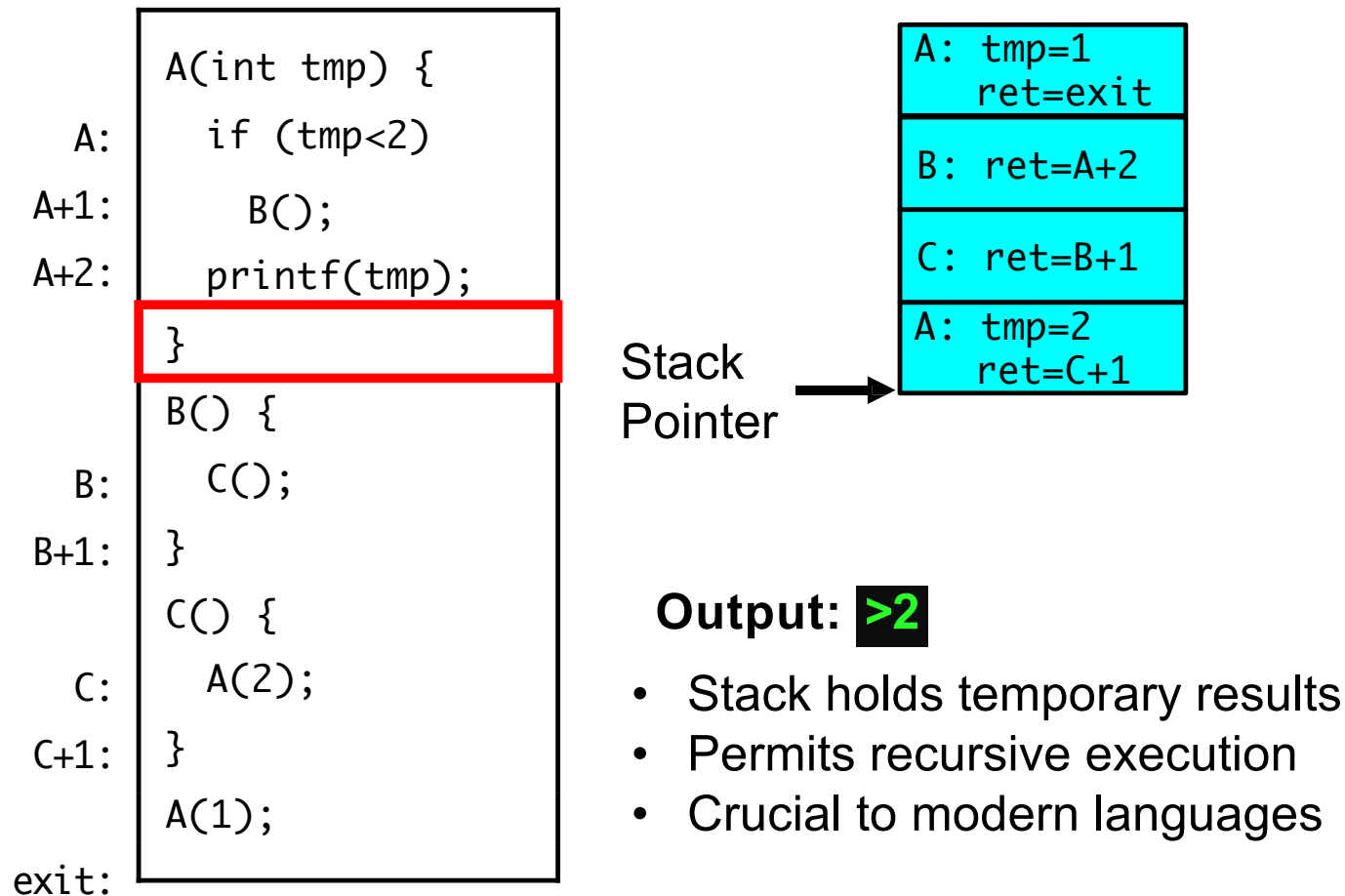
Execution stack example



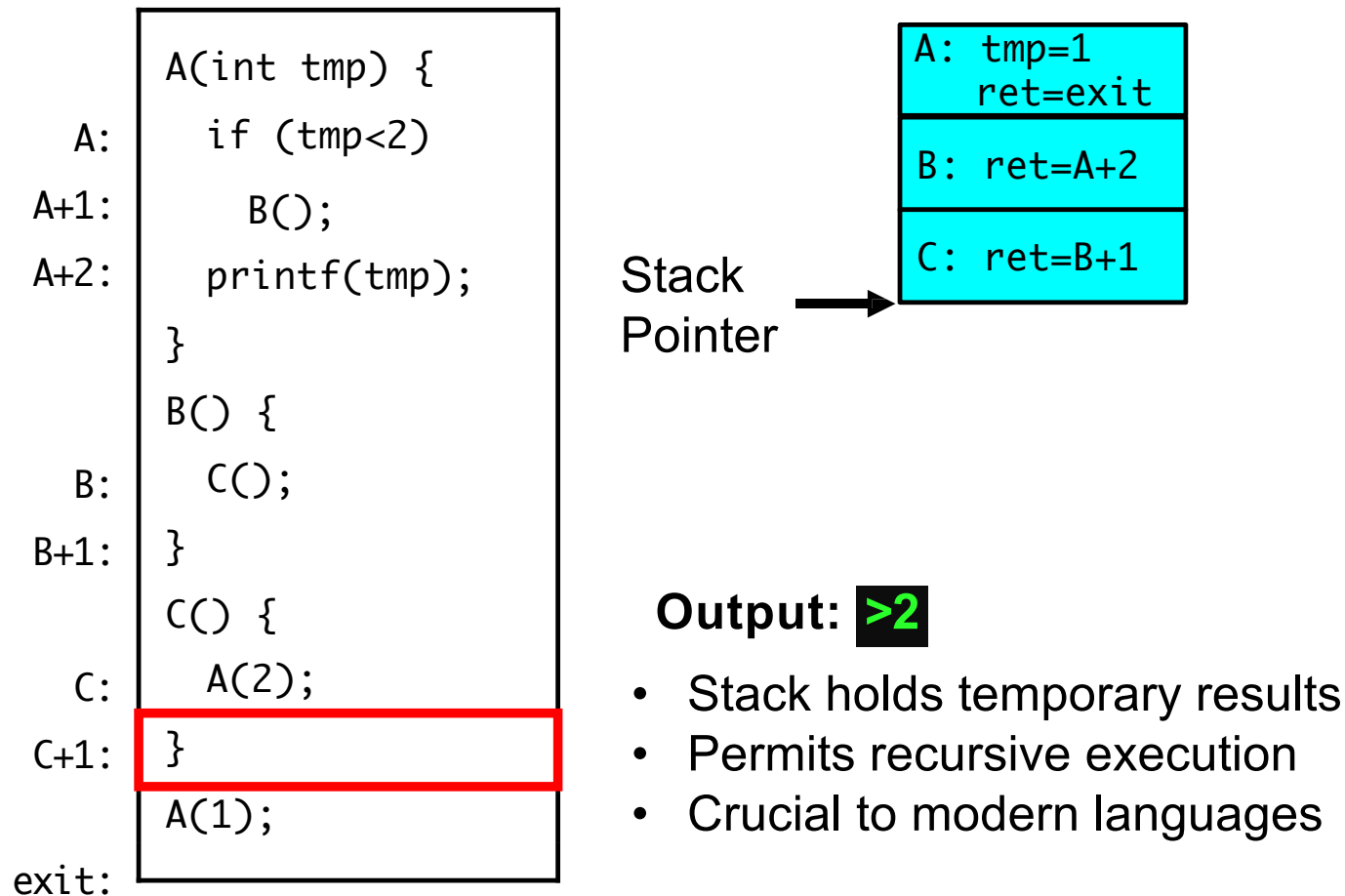
Execution stack example



Execution stack example



Execution stack example



Execution stack example

```

A:  A(int tmp) {
A+1:     if (tmp<2)
A+2:     B();
      printf(tmp);
      }
      B() {
B:      C();
B+1:    }
      C() {
C:      A(2);
C+1:    }
      A(1);
exit:

```

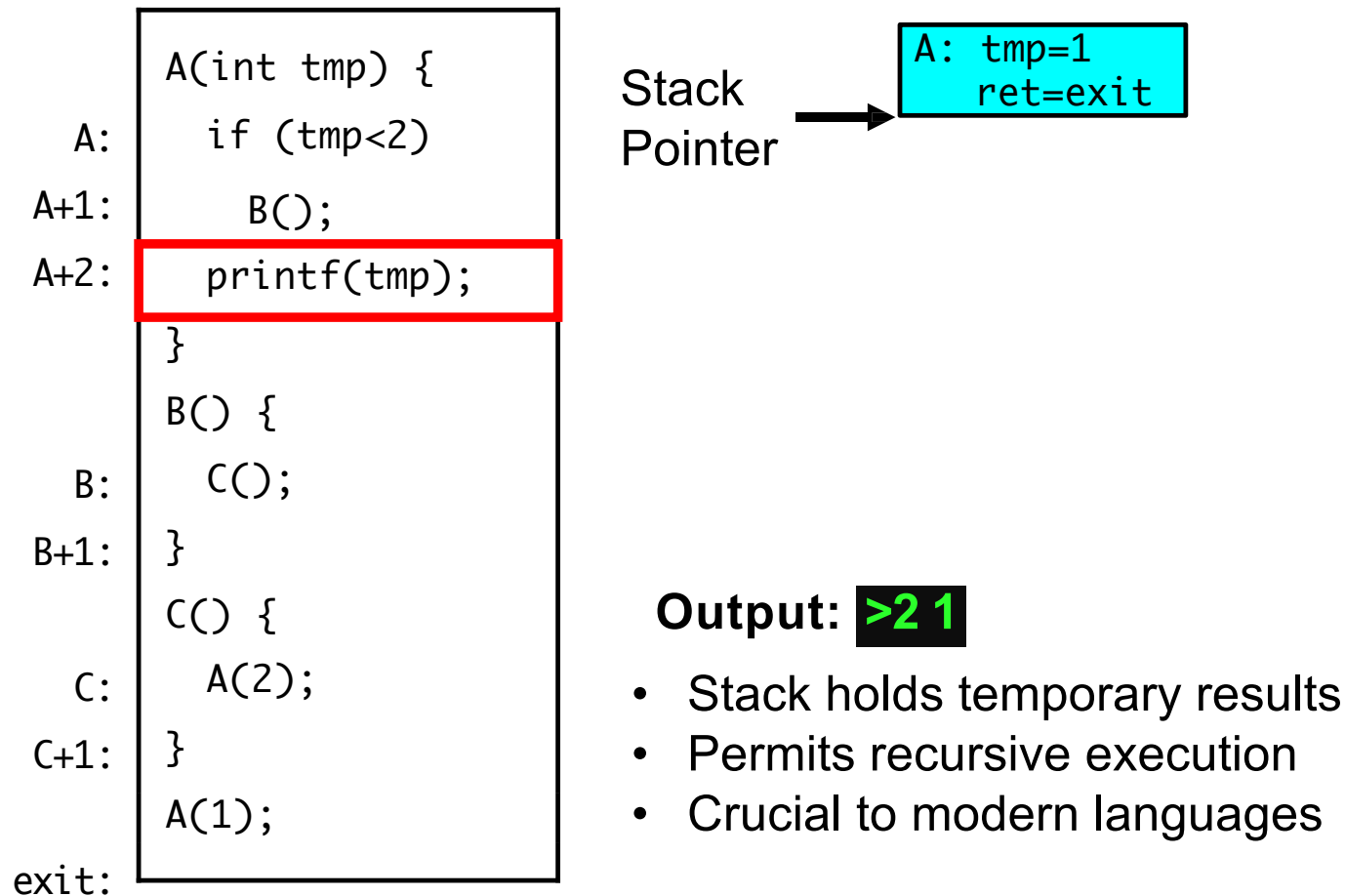
Stack
Pointer

A: tmp=1 ret=exit
B: ret=A+2

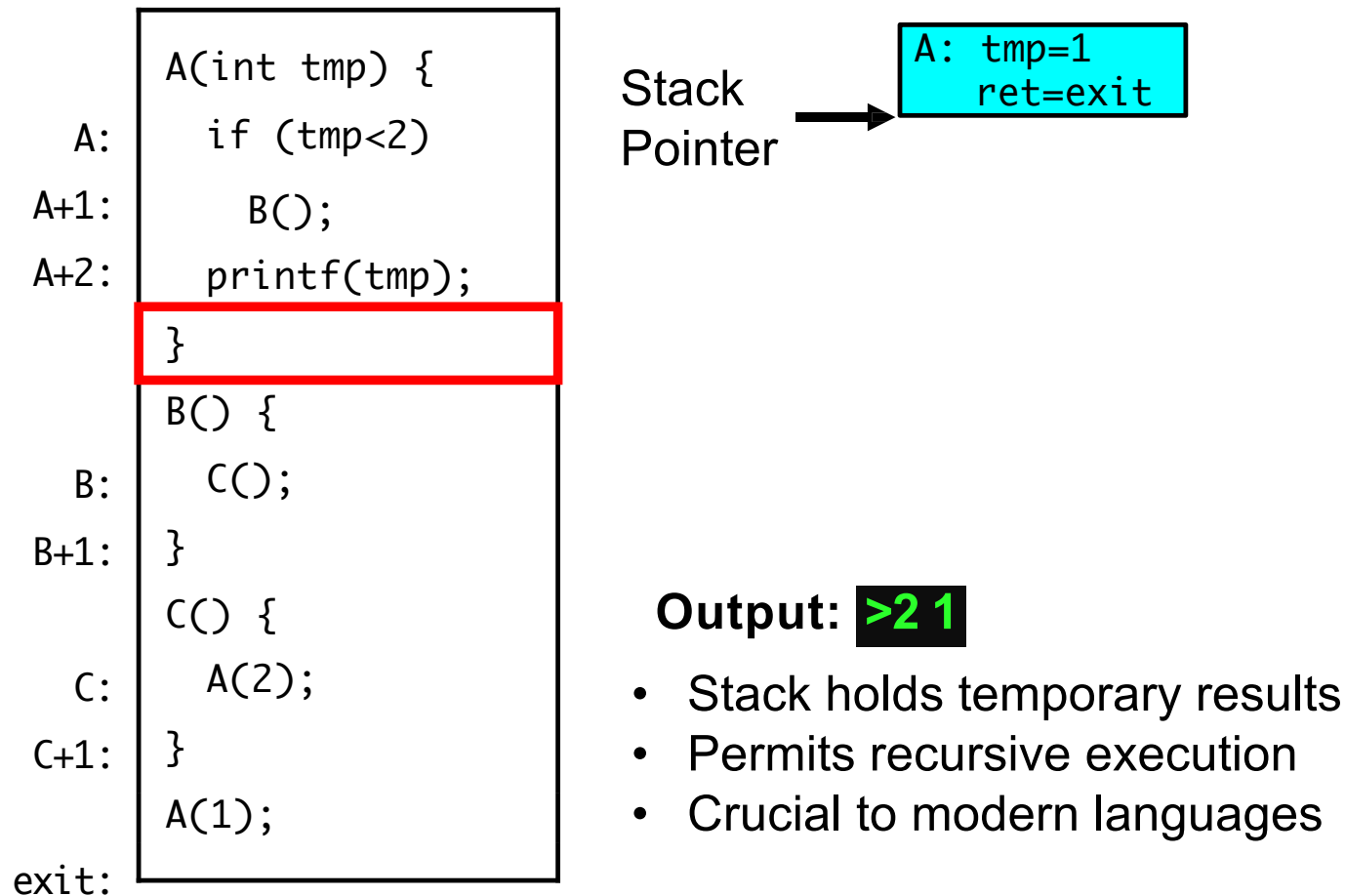
Output: >2

- Stack holds temporary results
- Permits recursive execution
- Crucial to modern languages

Execution stack example



Execution stack example



Execution stack example

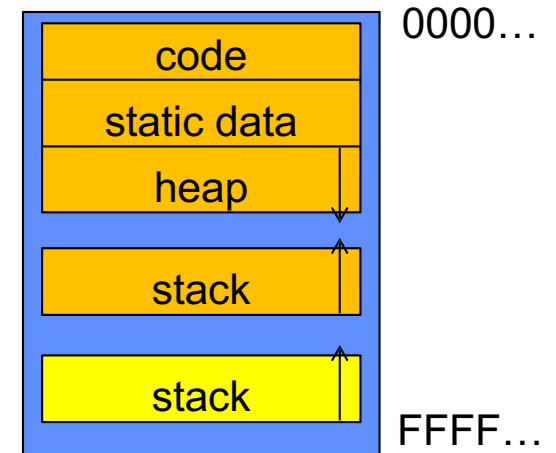
```
    A(int tmp) {  
A:      if (tmp<2)  
A+1:      B();  
A+2:      printf(tmp);  
      }  
      B() {  
B:      C();  
B+1:    }  
      C() {  
C:      A(2);  
C+1:    }  
      A(1);  
exit:    
```

Output: >2 1

- Stack holds temporary results
- Permits recursive execution
- Crucial to modern languages

Memory layout with two threads

- Two sets of CPU registers (where is it?)
- Two sets of Stacks
- Issues:
 - How do we position stacks relative to each other?
 - What maximum size should we choose for the stacks?
 - What happens if threads violate this?
 - How might you catch violations?



Conclusion

- Threads are the OS unit of concurrency
 - Abstraction of a virtual CPU core
 - Can use `pthread_create`, etc., to manage threads within a process
 - They share data → need synchronization to avoid data races
- Processes consist of one or more threads in an address space
 - Abstraction of the machine: execution environment for a program
 - Can use `fork`, `exec`, etc. to manage threads within a process
- We saw the role of the OS library
 - Provide API to programs
 - Interface with the OS to request services

Thanks